PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

PropSight: A Financial Monitoring and Reporting System for Data-Driven Rental Property Management

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Sonny Jr. E. Wagas - BSIT 3
Jr M. Marticio - BSIT 3
Marlon G. Villegas - BSIT 3
Kitche T. Camral - BSIT 3
Jasnine Feb C. Temillar - BSIT 3

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INTRODUCTION

1.1 Background of the Study

Rental property ownership is one of the most common forms of small-scale business enterprise in the Philippines. In cities and municipalities across the country, many families and individuals earn income by leasing out rooms, apartments, boarding houses, dormitories, and small commercial units. This is especially true in communities near universities, hospitals, commercial centers, and industrial zones where demand for affordable housing remains consistently high. For many property owners, rental income is a primary or supplemental livelihood that funds household needs, education, and healthcare.

Despite the widespread nature of rental property ownership, the management practices adopted by most small to medium-scale landlords remain informal and heavily dependent on manual methods. Many still rely on handwritten logbooks, carbon-copy receipts, and basic spreadsheets to keep track of monthly rental collections, outstanding tenant balances, utility expenses, repair costs, and other operational transactions. While these approaches are intended to work for someone managing only one or two units, the limitations become clear as operations grow. A landlord managing ten or more units across one or more properties faces a much greater volume and variety of financial transactions, making manual tracking increasingly unreliable and time-consuming.

The practical consequences of relying on fragmented manual financial management are well-documented in local research. Nercua et al. (2023), in their study of small-scale landlords in Davao City published in the International Journal of Industrial Management, found that financial challenges, including difficulty tracking income against expenses and managing outstanding balances, ranked among the most commonly cited operational constraints affecting landlords in the private rented sector. The study noted that the absence of organized financial records made it difficult for landlords to accurately assess the profitability of their properties, plan for maintenance costs, and make sound investment decisions. These findings closely mirror the experiences of property owners in other Philippine cities, including Iloilo, Cebu, and Metro Manila.

The problem is made worse by the fact that small to medium-scale property owners in the Philippines typically manage their properties without professional property management firms or dedicated accounting staff. Unlike large real estate developers, independent landlords perform all administrative, financial, and maintenance functions on their own, often while also holding other full-time employment. This places a heavy administrative burden on property owners and raises the risk of errors, oversights, and financial mismanagement that can negatively affect both the owner and their tenants.

From the perspective of the broader market, the Philippine rental housing sector has continued to grow. According to the Department of Human Settlements and Urban Development (DHSUD, 2022), the accumulated housing need in the Philippines reached 6.5 million units from 2017 to 2022, with a large proportion of urban households depending on the private rental market to meet their housing needs. The Philippine

Statistics Authority (2023) confirmed that the residential sector, including rental housing, remained a notable contributor to national economic output, with demand for rental units growing steadily in urbanizing areas. As the rental market expands, the need for more organized and efficient property management practices becomes increasingly essential.

Globally, the property management software market has grown rapidly in response to these needs. SkyQuest Technology (2024) projected that the global property management software market would surpass USD 11.54 billion by 2031, growing at a compound annual growth rate of 8.8%. This growth is driven largely by increasing demand for software-as-a-service platforms that automate routine property management tasks such as rent collection, tenant record maintenance, expense tracking, and financial reporting. While such solutions are widely available in developed markets, they are often too costly, complex, or infrastructure-intensive for small-scale landlords in developing countries like the Philippines.

The rapid growth of the property technology sector has also highlighted the potential of digital tools to transform property management at all scales. Naeem, Ahmed Rana, and Nasir (2023), in a systematic review published in Smart Construction and Sustainable Cities, identified real-time monitoring, data-driven decision-making systems, and information communication technologies as key enablers of more efficient property management. Their review noted that smaller operators in developing economies continue to lag behind in technology adoption due to cost barriers and the absence of tailored solutions. This gap underscores the need for accessible and locally relevant digital management tools designed specifically for small-scale property owners.

The challenge of managing financial records in small business contexts has also been studied outside the property sector. Okeke, Bakare, and Achumie (2024), in a case review published in the International Journal of Management and Entrepreneurship Research, found that data-driven financial systems significantly increase financial transparency and provide business owners with a competitive advantage by enabling faster and better-informed decision-making. The same study acknowledged that many small businesses face technical and financial barriers to adopting such systems, including limited budgets and the absence of industry-specific solutions. These findings apply directly to the situation of small-scale rental property owners in the Philippines.

To address these challenges, this study proposes PropSight, a Financial Monitoring and Reporting System for Data-Driven Rental Property Management. PropSight is a web-based system designed specifically for small to medium-scale rental property owners who manage their properties independently. The system consolidates all financial transactions, including rental income, operational expenses, maintenance costs, and tenant payment records, into a single, organized digital platform. By automating key financial processes such as report generation, payment tracking, and expense categorization, PropSight aims to give property owners accurate and timely financial insights. Through data visualization and dashboard analytics, the system supports informed decision-making, improves cash flow management, and supports owners' plans for the future with greater confidence.

The challenge of managing rental income and expenses is deeply rooted in the broader context of small-scale entrepreneurship in the Philippines. According to the

Department of Trade and Industry (DTI), micro, small, and medium enterprises (MSMEs) account for 99.5 percent of all registered businesses in the country and employ approximately 63 percent of the labor force. Rental property management, while not always formally registered as a business, functions in practice as a micro-enterprise for many Filipino families. As such, the financial management challenges faced by small-scale landlords closely mirror those documented in the broader MSME literature: informal record-keeping, reactive rather than proactive financial monitoring, and limited capacity for the kind of data-driven analysis that would support strategic growth and risk management.

The geographic context of this study is also relevant to understanding the problem. Iloilo City and the broader Western Visayas region have seen sustained population growth and urban expansion in recent years, driven by economic development, university enrollment growth, and migration from rural areas. This expansion has increased demand for rental housing, particularly boarding houses, dormitories, and apartment units near the University of the Philippines Visayas, Central Philippine University, West Visayas State University, and other major educational institutions. Property owners in these areas have responded to growing demand by expanding their rental portfolios, often without a corresponding upgrade in their financial management practices. The result is a growing mismatch between the scale and complexity of rental operations and the capacity of the informal tools used to manage them.

It is also worth noting that the shift toward digital financial management in the Philippine rental property sector has been uneven and largely driven by market forces rather than deliberate adoption strategies. A small number of larger property management companies operating in Metro Manila, Cebu, and other major urban centers have adopted commercial property management software packages, some of which are cloud-based platforms developed for the international market and adapted for local use. However, these platforms are typically designed for larger-scale operations and carry subscription fees that are prohibitive for the majority of small-scale landlords. The result is a bifurcated market in which technology-enabled property management is effectively accessible only to larger operators, while the majority of small-scale landlords continue to manage their finances using methods that have changed little in decades.

The COVID-19 pandemic, while primarily a public health crisis, also had notable implications for the Philippine rental property market and accelerated certain trends relevant to this study. The pandemic led to widespread displacement of urban renters, increased vacancy rates in areas dependent on student populations and office workers, and heightened financial stress for both landlords and tenants. For property owners, the pandemic period highlighted the inadequacy of manual financial management systems in tracking complex payment arrangements such as deferred rent, partial payments, and payment plans negotiated in response to tenant financial hardship. The experience of managing these arrangements using manual methods made the need for a more organized, digital approach to financial monitoring even more apparent to many small-scale landlords.

The post-pandemic recovery of the Philippine rental market has been accompanied by a broader national conversation about digital transformation across sectors. The Philippine government's Digital Philippines agenda, launched in 2022, emphasized the importance of digital tools for small businesses and identified access to affordable, sector-specific digital management tools as a priority area for intervention. While government programs have focused primarily on retail, food service, and agriculture, the rental property sector represents an essential but underserved area where targeted digital management tools have the potential to deliver considerable benefits in terms of financial transparency, operational efficiency, and property owner empowerment.

Against this backdrop, PropSight is designed not primarily as a technical solution to a record-keeping problem, but as a tool for financial empowerment for small-scale property owners in the Philippines. By giving property owners access to organized, accurate, and timely financial information about their rental businesses, PropSight aims to support improved financial decisions, reduce the stress and uncertainty associated with informal management practices, and ultimately contribute to the viability and sustainability of small-scale rental property businesses as a source of livelihood for Filipino families.

The name PropSight itself reflects this vision. The word "Prop" is derived from "property," indicating the system's domain focus, while "Sight" refers to visibility and insight: the ability to see clearly and to understand what one is seeing. PropSight is designed to give property owners clear sight of their financial position, enabling them to manage their properties with greater confidence and effectiveness. The tagline "Data-Driven Rental Property Management" further emphasizes the system's purpose of transforming raw financial data into actionable management insights, moving property owners from reactive management based on intuition to proactive management grounded in evidence.

1.2 Statement of the Problem

Despite the increasing number of rental properties and the growing dependence on rental income, many property owners continue to face difficulties in effectively managing and monitoring their financial transactions. These challenges often result in inefficiencies, inaccuracies, and limited financial insight. Many property owners still rely on manual methods such as notebooks, spreadsheets, or unorganized digital files, which can lead to errors, duplication of entries, and even data loss. Tracking rental payments, recording expenses, and monitoring outstanding balances can become complicated, especially when managing multiple properties or tenants at the same time. Inconsistent record-keeping has the potential to result in missed payments, overlooked expenses, or incorrect financial reporting, which can affect the overall financial condition of the property owner.

Furthermore, the absence of an integrated and centralized system makes it difficult for property owners to access real-time financial information whenever needed. This limits their ability to make timely and well-informed decisions regarding their investments and operations. Without proper tools, analyzing financial performance, identifying income trends, and detecting unnecessary expenses becomes challenging and time-consuming. Property owners could struggle to evaluate profitability, manage cash flow efficiently, and prepare accurate financial reports. In addition, the lack of automation increases the workload and the possibility of human error in handling financial data. Communication gaps between property owners and tenants regarding payments may also arise due to poor tracking systems. These issues can negatively affect overall property management, decision-making, and long-term financial stability. Therefore, there is a strong need for a reliable, accurate, and user-friendly system that can simplify financial tracking, improve data accuracy, provide real-time monitoring, and support improved decision-making for property owners.

In addition to these challenges, there is also a lack of systematic financial reporting tools specifically designed for small to medium-scale property owners. Most existing approaches do not provide automated generation of financial summaries such as monthly income reports, expense breakdowns, and payment status reports. As a result, property owners are often required to manually compute and compile financial data, which increases the risk of inconsistencies and delays in reporting.

Another concern is the limited capability of current manual and semi-digital methods to support data-driven decision-making. Without structured data analysis and visualization, property owners are unable to easily identify trends in rental income, occupancy performance, or recurring expenses. This lack of analytical capability prevents them from making proactive financial decisions that could improve profitability and operational efficiency.

Moreover, there is minimal integration between tenant management and financial monitoring in existing systems used by small-scale landlords. This separation of processes often results in fragmented information, where tenant records and financial records are stored independently, making reconciliation more difficult. A unified system that connects tenant information with financial transactions is necessary to ensure consistency and accuracy across all records.

Lastly, many available solutions in the market are either too complex, too expensive, or designed for large-scale property management companies. This creates a gap in the availability of affordable and user-friendly systems tailored specifically for independent landlords and small property owners. This gap highlights the need for a system like PropSight, which is designed to be accessible, simple to use, and focused on essential financial monitoring and reporting functions.

Specifically, this study aims to address the following problems:

1. **Financial records of rental income and expenses are frequently managed manually or through fragmented systems, leading to disorganized data handling.** This makes it difficult to maintain complete and reliable records, especially when dealing with multiple tenants and properties. Many property owners still rely on spreadsheets, notebooks, or separate applications to manage their financial data. This scattered approach often results in missing or duplicated entries. Over time, it becomes harder to track historical transactions accurately. Important financial information may be stored in different locations, making retrieval time-consuming. This lack of organization increases the risk of confusion in financial reporting. It also makes it challenging to monitor overall property performance effectively. Without a unified system, consistency in record-keeping is difficult to achieve. As a result, financial management becomes inefficient and prone to errors. This issue becomes more pronounced as property owners handle increasing volumes of financial data from multiple tenants, where

manual tracking systems are no longer sufficient to ensure accuracy and consistency. The continued reliance on fragmented recording methods also makes it difficult to consolidate financial information into a single reliable source for analysis and review.

In addition, the lack of a unified approach to financial record management limits the ability to maintain consistent documentation of rental income and expenses over time. This inconsistency directly affects the reliability of financial records, especially when comparing historical data or verifying past transactions.

Furthermore, fragmented financial handling creates difficulties in ensuring that all income and expense entries are properly recorded and accounted for within a single system. This often leads to incomplete financial visibility, making it harder for property owners to maintain accurate and organized records necessary for effective property management.

2. **Property owners encounter difficulties in generating accurate, consistent, and timely financial reports. As a result, decision-making is often delayed or based on incomplete or outdated information.** Preparing reports manually can take a significant amount of time and effort. This delay prevents property owners from quickly understanding their financial status. In many cases, reports may contain errors due to manual calculations. Inconsistent data sources can also affect the reliability of financial summaries. Property owners may struggle to compare income and expenses across different time periods. This makes it difficult to evaluate profitability accurately. Without timely reports, necessary financial decisions may be postponed. Outdated information can lead to poor planning and missed opportunities. Therefore,

efficient reporting tools are essential for improved financial management. This issue is further intensified when financial records are not properly consolidated, as scattered data sources make it difficult to ensure that all relevant transactions are included in the report generation process. When information is stored across multiple files or platforms, property owners may unintentionally overlook necessary financial entries, resulting in incomplete reporting outputs.

In addition, the absence of standardized reporting formats contributes to inconsistencies in how financial data is presented and interpreted. Without a uniform structure for financial reports, it becomes challenging to maintain clarity and ensure that comparisons across different reporting periods are accurate and meaningful.

Furthermore, delays in generating financial reports reduce the ability of property owners to respond promptly to financial concerns such as overdue payments, unexpected expenses, or declining profitability. This lack of timely insight limits their capacity to make proactive decisions that could improve financial outcomes.

Lastly, reliance on manual report preparation increases the likelihood of human error, especially when handling large volumes of financial data. Even minor inaccuracies in computation or data entry can significantly affect the reliability of financial reports, ultimately impacting the quality of decision-making based on those reports.

3. **There is a lack of real-time monitoring of rental payments, outstanding balances, and operational expenses.** This limits the ability of property owners to promptly track financial activities and respond to issues such as late payments. Many current systems only provide updates after manual entry or periodic checks. This delay

prevents immediate awareness of financial changes. Property owners may not know right away if tenants fail to pay on time. Late detection of payment issues can affect cash flow and budgeting. Operational expenses may also go unnoticed until reviewed later. Without real-time updates, financial tracking becomes less reliable. It becomes harder to manage multiple tenants efficiently. Immediate action on financial issues is often not possible. This highlights the need for an automated monitoring system. This issue becomes more critical in situations where multiple tenants and properties are being managed simultaneously, as delayed updates can quickly lead to accumulated discrepancies in financial records. Without continuous monitoring, small payment delays or unnoticed expenses may compound over time, making it more difficult to reconcile accounts accurately.

In addition, the absence of real-time monitoring reduces the ability of property owners to maintain an up-to-date overview of their financial position. This lack of immediacy affects not only payment tracking but also the monitoring of outstanding balances, which may remain unresolved for extended periods without timely intervention.

Furthermore, delayed visibility of operational expenses can lead to ineffective budget control, as property owners are unable to react promptly to unexpected costs or financial deviations. This may result in poor allocation of funds and reduced financial stability over time.

Lastly, the lack of automated and real-time updates limits the ability to generate instant financial insights needed for quick decision-making. This gap in monitoring capability highlights the importance of implementing a system that can continuously update financial records as transactions occur, ensuring that property owners always have access to current and reliable financial information.

4. **Financial data is not presented in a structured and analytical format, limiting its usefulness for informed and data-driven decision-making.** Consequently, property owners may struggle to identify financial trends and optimize their management strategies. Raw financial data is often difficult to interpret without proper organization. Many systems fail to convert data into meaningful insights. Property owners may find it challenging to understand patterns in income and expenses. Without proper analysis tools, decision-making becomes based on assumptions. This can lead to ineffective financial strategies. Identifying profitable and unprofitable properties becomes harder. Trends such as seasonal income changes may go unnoticed. Opportunities for cost reduction may also be missed. Structured data presentation is essential for improved understanding. Analytical tools help improve long-term financial planning. This limitation becomes more evident when financial information is accumulated over long periods, as unstructured data makes it difficult to perform meaningful comparisons across different time frames. Without a structured format, historical financial data cannot be efficiently analyzed to support forecasting or performance evaluation.

In addition, the absence of analytical presentation tools such as summaries, breakdowns, or categorized reports reduces the ability of property owners to interpret complex financial information. This prevents them from easily distinguishing between essential and non-essential expenses or identifying the primary sources of rental income.

Furthermore, the lack of structured financial visualization contributes to difficulty in recognizing hidden patterns within financial activities. Important indicators such as recurring expenses, inconsistent payment behavior, and gradual changes in profitability may remain unnoticed without proper analytical representation.

Lastly, the inability to transform raw financial data into actionable insights limits the overall effectiveness of financial planning. Without structured analysis, property owners are less equipped to make evidence-based decisions, resulting in less strategic and less optimized management of rental properties.

5. **Manual processes increase the risk of human errors, data inconsistencies, and reduced transparency in financial management.** This can lead to discrepancies in records and decreased trust in the accuracy of financial information. When data is entered manually, mistakes are more likely to occur. Simple errors in numbers can affect overall financial reports. Inconsistencies may arise when different methods are used for recording data. These issues make it difficult to verify financial accuracy. Transparency is reduced when records are unclear or incomplete. Property owners may find it hard to justify financial results. Tenants and stakeholders may also lose trust in the system. Reconciliation of financial records becomes time-consuming. Errors may accumulate over time if not corrected. Therefore, automated and structured systems are needed to improve accuracy and trust. This

issue becomes more critical as the volume of financial transactions increases, since manual handling of large datasets significantly amplifies the likelihood of unnoticed errors. As records grow over time, even minor inaccuracies can accumulate and lead to substantial discrepancies that are difficult to trace and correct.

In addition, manual processes often lack standardized validation mechanisms, which means that incorrect entries may go undetected during data input. Without automated checks or verification features, financial data integrity becomes heavily dependent on human attention, increasing vulnerability to oversight and inconsistency.

Furthermore, the absence of a unified system for financial recording reduces overall transparency, as it becomes difficult to track how and when data entries were made. This lack of traceability limits accountability and makes it challenging to audit financial activities effectively when discrepancies arise.

Lastly, reliance on manual processes reduces the efficiency of financial reconciliation, as property owners must manually compare and verify multiple records across different sources. This not only consumes more time but also increases the risk of overlooking errors, further affecting the reliability of financial reporting and overall trust in the system.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to design, develop, and evaluate PropSight, a Financial Monitoring and Reporting System for rental property management, with the aim of helping small to medium-scale property owners in the Philippines efficiently track income and expenses, produce accurate financial reports, monitor tenant payment statuses in real time, and make well-informed, data-driven decisions about the operation and growth of their rental properties. In addition, this study seeks to contribute to the improvement of rental property management practices by promoting the use of centralized and automated financial systems that reduce reliance on manual processes. Through the implementation of PropSight, property owners may experience enhanced data accuracy, improved financial transparency, and more efficient handling of rental-related transactions. Furthermore, the system is expected to support improved organizational control over financial records, enabling users to maintain consistency in reporting and gain clearer insights into overall property performance. It also aims to reduce common operational challenges such as delayed financial updates, incomplete records, and difficulty in reconciling tenant payments and expenses. By addressing these issues, the system may help property owners minimize financial risks associated with human error and fragmented record-keeping. Moreover, the study encourages the adoption of data-driven practices in small to medium-scale property management by providing tools that support real-time monitoring, structured reporting, and simplified financial analysis. Ultimately, PropSight is intended to serve as a practical solution that enhances efficiency, supports improved decision-making, and improves the overall financial management experience of property owners in a digital environment.

1.3.2 Specific Objectives

Specifically, this study aims to:

1. The system aims to develop a centralized platform that allows rental property owners to efficiently record and manage financial data. It will provide a unified space where all rental income and operational expenses can be stored and accessed. By consolidating financial information, the system reduces the need for multiple tools or manual record-keeping. This approach improves accuracy and organization in handling financial transactions. Ultimately, it enhances the overall efficiency of property management operations. In addition, the system seeks to ensure that financial data is consistently structured and properly categorized to support easier retrieval and review of records. This includes organizing transactions in a way that allows property owners to quickly identify specific income sources, expense types, and payment histories without confusion or data fragmentation. Furthermore, the centralized platform is expected to minimize redundancy in data entry and reduce inconsistencies that often occur when using separate or manual systems. By improving data organization and accessibility, the system supports more reliable financial tracking and contributes to a more efficient overall management of rental properties.

2. The study seeks to design and implement a feature that enables the generation of accurate and timely financial reports. These reports will help property owners monitor income, expenses, and overall financial performance. The system will ensure that data is processed systematically to avoid discrepancies. It will also allow users to generate reports whenever needed for decision-making purposes. As a result, property owners can gain more accurate insights into the financial status of their rental properties. In addition, the reporting feature aims to support different types of financial summaries that may be useful for property evaluation, such as periodic income

summaries and categorized expense breakdowns. This will help users improve their understanding on how their financial resources are being allocated across different properties or tenants. Furthermore, the system is designed to improve the consistency of reporting outputs by using standardized data processing methods, ensuring that generated reports remain uniform regardless of the time or frequency of access. This also supports easier comparison of financial performance across different periods, allowing property owners to identify changes and evaluate overall trends more effectively.

3. The system will provide a real-time monitoring mechanism for tracking rental payments and financial activities. It will allow users to view updated records of tenant payments as they occur. Outstanding balances and operational costs will also be monitored continuously. This feature ensures that property owners are always informed about their financial standing. Consequently, it supports timely actions and improved financial control. In addition, the real-time monitoring feature aims to reduce delays in identifying financial issues such as missed or late payments, allowing property owners to respond more quickly to potential problems. It also improves financial awareness by ensuring that updates are immediately reflected in the system without the need for manual checking or periodic review. Furthermore, this continuous monitoring helps maintain consistency between recorded transactions and actual financial activity, reducing the likelihood of unnoticed discrepancies. By providing always-updated information, the system enhances overall reliability in financial oversight and supports more proactive property management practices.

4. The system is designed to organize and present financial data in a structured and analytical format. Information will be arranged in a way that is easy to interpret and understand. It will incorporate data visualization and summaries to highlight key financial trends. This structured presentation helps users analyze performance more effectively. In turn, it supports informed and data-driven decision-making. In addition, the structured presentation of financial data aims to improve the clarity and accessibility of complex financial information, especially when dealing with multiple tenants, properties, and transaction types. By organizing data into categorized and meaningful formats, users can more easily distinguish between different financial elements such as income sources, expense classifications, and payment statuses. This reduces cognitive load and allows property owners to interpret financial information with greater efficiency.

Furthermore, the inclusion of analytical presentation methods helps ensure that financial data is not only displayed but also made useful for evaluation purposes. This allows property owners to identify inconsistencies, compare financial performance across different periods, and assess overall profitability with greater accuracy. As a result, decision-making becomes more evidence-based rather than reliant on raw or unprocessed data.

Lastly, this feature also supports more efficient long-term financial planning by enabling users to recognize patterns and trends that may not be immediately visible in unstructured data. Through improved organization and analytical presentation, the system enhances the overall usefulness of financial information and strengthens the ability of property owners to make strategic management decisions.

5. The system aims to minimize errors, inconsistencies, and lack of transparency associated with manual financial management processes. Automation will reduce the risk of human error in recording and calculating financial data. It will also ensure consistency in data entry and reporting. By maintaining accurate and transparent records, the system builds trust and reliability. Overall, it improves the integrity and efficiency of financial management practices. In addition, the system seeks to address issues that arise from repetitive manual handling of financial information, where even small inaccuracies can accumulate over time and affect overall record reliability. Through automated processing, the likelihood of duplicate entries, missed transactions, and incorrect computations is significantly reduced, ensuring that financial records remain dependable and up to date.

Furthermore, the implementation of a structured digital system enhances accountability by providing clearer traceability of financial records. This makes it easier to identify when and how data entries were made, which supports verification and correction processes when discrepancies occur. As a result, property owners gain greater confidence in the accuracy and completeness of their financial data.

Lastly, by reducing reliance on manual processes, the system supports more efficient reconciliation of financial records. This helps ensure that all income, expenses, and payment activities are properly aligned and consistently reflected within the system, strengthening overall financial control and reducing the risk of unnoticed errors affecting long-term financial outcomes.

1.4 Scope and Delimitations

Scope of the Study

This study focuses on the design and development of PropSight: A Financial Monitoring and Reporting System for Data-Driven Rental Property Management, a web-based application intended to assist rental property owners in managing their financial transactions in a more organized, efficient, and accurate manner.

The system is specifically designed for small- to medium-scale property owners such as those managing apartments, boarding houses, dormitories, and small commercial rental spaces. It aims to centralize financial records and improve monitoring of rental-related income and expenses.

The study covers the development of a centralized platform that consolidates financial information related to rental property operations into a single, organized system. Through this platform, property owners can maintain records of rental income, operational expenses, tenant payment activities, and other financial transactions associated with the management of their rental properties. By replacing fragmented and manual record-keeping practices, the system seeks to improve the accessibility, accuracy, and consistency of financial information.

In addition, the study focuses on providing property owners with tools that support financial monitoring and reporting activities. The system is designed to facilitate the tracking of payment statuses, monitoring of outstanding balances, generation of financial summaries, and presentation of financial data in a structured format that is easier to analyze and interpret. These capabilities are intended to assist property owners in evaluating the financial performance of their rental operations and making more informed management decisions.

The scope of the study also includes the design of user interfaces, database structures, and system processes necessary to support the recording, storage, retrieval, and presentation of financial information. Particular emphasis is placed on ensuring that the system remains accessible and practical for property owners who may have limited experience with specialized financial management software. Through the integration of these components, PropSight aims to provide a reliable and user-friendly solution for rental property financial management.

The scope of the system includes the following features:

- Web-based recording and management of rental income and operational expenses
- Monitoring of tenant payments, due dates, and outstanding balances
- Automated generation of financial reports such as income summaries, expense breakdowns, and overall financial performance reports
- Real-time viewing of financial transactions and account status

Organized presentation of financial data for easier analysis and interpretation

- User authentication for secure access of property owner accounts
- A responsive and user-friendly web interface accessible through modern web browsers

The system will be developed using web technologies and will utilize a centralized database to ensure data consistency, accessibility, and efficient management of financial records. It is intended to support property owners in improving financial transparency and enabling data-driven decision-making.

Delimitations of the Study

Despite its intended functionality and benefits, the system has the following limitations:

- The system is limited to financial monitoring and reporting and does not include full property management features such as tenant communication, messaging systems, or lease contract generation.
- It is dependent on user input; therefore, the accuracy and reliability of financial records depend on the correctness of the data entered by the property owner or user.
- The system is designed primarily for small- to medium-scale rental property operations and may not fully support large-scale enterprise-level property management systems.
- As a web-based application, the system requires an internet connection to access and use its features, which may affect usability in areas with poor or unstable connectivity.
- The system does not include advanced technologies such as artificial intelligence-based financial forecasting or predictive analytics.
- Security features are limited to basic authentication and data protection mechanisms and may not meet advanced enterprise cybersecurity standards.

The system is designed to manage financial information related to rental property operations and is not intended to replace professional accounting software or the services of certified accounting professionals. Financial reports generated by the system are intended for monitoring and decision-support purposes and are not official accounting or legal documents.

The system's performance and responsiveness may vary depending on the hardware specifications of the device being used, the speed and stability of the internet connection, and the volume of records stored in the database. As the number of transactions and users increases, additional optimization and infrastructure improvements may be required to maintain optimal performance.

Furthermore, the study focuses on the development and evaluation of the system within a controlled academic and testing environment. Long-term operational performance, large-scale deployment scenarios, and extended real-world usage beyond the study period are not included within the scope of the system evaluation.

The system supports only the functionalities identified during the requirements-gathering phase of the study. Requests for additional features or specialized business processes that emerge after development are not included in the current version of PropSight and may be considered for future enhancements and subsequent research.

1.5 Significance of the Study

The proposed system, PropSight: A Financial Monitoring and Reporting System for Data-Driven Rental Property Management, is expected to provide benefits to the following individuals and groups:

Property Owners. The primary beneficiaries of this study are property owners who manage rental spaces such as apartments, boarding houses, dormitories, and small commercial units. The system will assist them in efficiently monitoring rental income, tracking operational expenses, and managing outstanding payments in a centralized platform. By replacing manual and fragmented record-keeping methods, PropSight reduces the likelihood of human errors, data inconsistencies, and missing records. It also saves time in generating financial reports and provides accurate, organized, and real-time financial information. As a result, property owners can make more informed and data-driven financial decisions, improving overall business efficiency and profitability.

Tenants. Although the system is primarily designed for financial monitoring from the property owner's perspective, tenants will also indirectly benefit from its implementation. With improved tracking and organized record-keeping of payments, tenants may experience more transparent and accurate billing systems. This reduces misunderstandings or disputes regarding rental balances, due dates, and payment history. Ultimately, the system promotes clearer communication and accountability between tenants and property owners.

Future Researchers. This study can serve as a valuable reference for future researchers who intend to explore topics related to financial monitoring systems, rental property management systems, and data-driven decision-making tools. It may provide insights into system design, development methodologies, and functional requirements that can be adapted or enhanced in future studies. Additionally, it can serve as a foundation for improving or expanding similar systems in the context of small- to medium-scale business operations.

System Developers and IT Practitioners. This study offers a comprehensive framework for the design, development, and implementation of financial monitoring systems within web-based property management environments, providing valuable insights into the processes, methodologies, and technical considerations involved in creating efficient and reliable digital solutions for financial management. It can serve as a practical guide and reference for developers, programmers, system analysts, and IT practitioners who intend to create similar web-based property management applications, as it presents relevant concepts, development strategies, and implementation techniques that can be adapted and applied to various organizational settings and operational requirements. Moreover, it demonstrates practical approaches to integrating essential financial management functionalities, including financial tracking, automated reporting, transaction recording, budget monitoring, and interactive data visualization features, which are crucial for improving the accuracy, accessibility, and transparency of financial information within a property management system. Through the findings and system design presented in this study, developers can also gain valuable insights into enhancing system usability, scalability, maintainability, and overall functionality, enabling them to develop applications that effectively address real-world challenges, accommodate future growth and technological advancements, and provide a more efficient and user-friendly experience for both administrators and end users in property management operations.

Academic Institution. For PHINMA University of Iloilo, this study demonstrates the applied, practical dimension of information technology education. It shows that BSIT students pursuing the Business Analytics Management Track can design and develop technology-driven solutions to real-world business problems and contribute to the institution's portfolio of student-led research and development projects.

1.6 Definition of Terms

The following terms are defined operationally as they are used within the context of this study:

PropSight. The proposed web-based Financial Monitoring and Reporting System developed in this study, designed to help small to medium-scale rental property owners track income and expenses, monitor tenant payment statuses, generate financial reports, and make data-driven management decisions. By centralizing financial and tenant-related information into a single platform, the system reduces the challenges associated with manual record-keeping and improves the accuracy and accessibility of financial data. The system also provides real-time monitoring and reporting capabilities that enable property owners to quickly identify financial trends, track outstanding payments, and evaluate the overall performance of their rental properties. In addition, PropSight supports more informed and strategic decision-making by presenting financial information through organized reports and visual summaries, allowing users to manage their properties more effectively and sustainably.

Furthermore, PropSight serves as a centralized information management platform that enables property owners to maintain complete and organized records of their rental operations within a single database environment. The system integrates various financial processes, including transaction recording, payment monitoring, expense management, and report generation, reducing the need to maintain separate spreadsheets, notebooks, or manual records. Through the consolidation of these functions, users can access accurate and up-to-date financial information whenever needed.

PropSight also functions as a decision-support tool by transforming raw financial data into meaningful information that can be used to assess the financial condition of rental properties. By providing structured reports, historical transaction records, and dashboard-based summaries, the system assists users in identifying patterns, evaluating income and expense performance, and monitoring the overall profitability of their rental operations. These capabilities contribute to improved financial planning, resource allocation, and operational management.

Furthermore, the system is designed to promote transparency, accountability, and efficient record management by ensuring that financial information is stored systematically and can be retrieved quickly for review and analysis. As a web-based solution, PropSight provides flexibility and accessibility, allowing authorized users to manage and monitor rental property finances through a centralized digital platform.

Through these functions, the system supports the modernization of rental property management practices and encourages the adoption of technology-driven approaches to financial administration.

Rental Property Management. Any residential or commercial real estate asset, including apartments, boarding houses, dormitories, and small commercial spaces, that is leased or rented to tenants in exchange for regular monetary payment. These properties serve as a source of income for property owners and require proper management to ensure efficient operations and consistent revenue generation. Rental properties often involve various financial transactions, such as rent collection, maintenance expenses, utility payments, and other operational costs that must be monitored regularly. Effective management of rental properties helps property owners maintain profitability, improve tenant satisfaction, and make informed decisions regarding property investments and maintenance.

Furthermore, rental properties require continuous administrative and financial oversight to ensure that rental operations remain organized, sustainable, and profitable over time. Property owners are responsible for managing tenant occupancy, monitoring rental payments, addressing maintenance concerns, and overseeing day-to-day operational activities. These responsibilities generate large amounts of financial and operational information that must be recorded and managed accurately to support effective property administration.

In addition, the financial performance of a rental property is influenced by various factors, including occupancy rates, rental income, maintenance expenditures, utility costs, and other recurring operational expenses. Monitoring these factors enables property owners to evaluate the profitability of their properties, identify potential financial issues, and implement appropriate management strategies. Accurate monitoring also helps reduce the risk of missed payments, unrecorded expenses, and financial inconsistencies that may negatively affect business performance.

Moreover, rental properties represent long-term investments that require informed decision-making regarding budgeting, maintenance planning, property improvements, and future expansion opportunities. Access to organized financial records and reliable operational information allows property owners to make strategic decisions that contribute to the stability, growth, and overall success of their rental property business. As a result, effective rental property management plays a significant role in ensuring both operational efficiency and financial sustainability.

Financial Records. Systematic digital documentation of all monetary transactions associated with the operation of rental properties, including records of rental income received, operational expenses incurred, maintenance costs, and outstanding balances. They provide a complete and organized history of financial activities, ensuring accuracy and transparency in property management. These records help property owners monitor cash flow, track financial performance, and identify trends that may affect profitability. They also support budgeting and financial planning by providing reliable data for forecasting future income and expenses. Additionally, maintaining accurate financial records facilitates compliance with accounting standards, simplifies audit processes, and enables informed decision-making regarding property operations and investments.

Furthermore, financial records serve as an essential source of information for evaluating the overall financial health of rental property operations. By maintaining accurate and up-to-date records of all financial transactions, property owners can gain a clearer understanding of revenue generation, expense patterns, and net financial performance over specific periods. This information enables them to assess whether their properties are meeting financial objectives and identify areas that may require corrective action or operational improvement.

In addition, well-maintained financial records promote accountability and transparency by ensuring that all monetary activities are properly documented and can be verified when necessary. Organized records reduce the likelihood of errors, omissions, and duplicate entries, while also making it easier to trace transactions and resolve discrepancies. The availability of accurate financial information contributes to more effective financial control and strengthens confidence in the reliability of reported financial data.

Moreover, digital financial records improve the efficiency of information storage, retrieval, and analysis compared to traditional paper-based methods. Through electronic record management, users can quickly access historical transaction data, generate financial summaries, and review trends that support strategic planning and resource allocation. The ability to organize and analyze financial information systematically allows property owners to respond more effectively to changing operational and financial conditions.

Financial records also provide a valuable foundation for performance measurement and long-term business planning. By comparing financial data across different reporting periods, property owners can evaluate growth, monitor profitability, and determine the effectiveness of management strategies. As a result, accurate financial records are not only tools for documentation but also critical resources for supporting informed decision-making, operational efficiency, and sustainable property management practices.

Real-Time Monitoring. The capability of a system to capture, process, and display financial data such as payment statuses and outstanding balances immediately as transactions occur, without significant delay between data entry and data availability. It provides users with up-to-date information, allowing them to track financial activities and property operations as they happen. This feature helps property owners and administrators quickly identify issues, monitor payment compliance, and respond to changes in financial conditions promptly. By ensuring that information is continuously updated, real-time monitoring improves decision-making, enhances operational efficiency, and supports accurate financial management.

Furthermore, real-time monitoring supports greater visibility into financial and operational activities by ensuring that newly entered information is immediately reflected throughout the system. This enables users to access current data without waiting for manual updates or periodic report generation. Through continuous data availability, property owners can maintain a more accurate oversight of rental operations, detect irregularities more quickly, and ensure that financial records remain accurate and up to date. As a result, real-time monitoring contributes to improved responsiveness, improved resource management, and more effective control over property-related financial activities.

Data-Driven Decision-Making. A management approach in which decisions are informed by the systematic analysis of organized data rather than intuition or guesswork. In PropSight, this refers to using financial reports, dashboards, and analytical summaries to guide property management strategies.By relying on accurate and up-to-date information, property owners and administrators can make more objective and informed decisions regarding rental operations, budgeting, and resource allocation. This approach helps identify trends, evaluate performance, and uncover opportunities for improvement. It also reduces the risk of errors and uncertainty by providing evidence-based insights that support effective planning and long-term business growth.

Furthermore, data-driven decision-making promotes a more systematic and objective approach to property management by encouraging decisions based on measurable information rather than assumptions or personal judgment alone. Through the analysis of historical and current financial data, property owners can improve their understanding of the factors affecting the performance of their rental properties and make decisions supported by reliable evidence. This approach improves the accuracy of planning activities and helps managers respond more effectively to operational and financial challenges.

In addition, data-driven decision-making enables property owners to compare financial performance across different reporting periods, evaluate the effectiveness of management strategies, and monitor progress toward financial goals. By examining trends and patterns in rental income, expenses, occupancy levels, and payment behavior, users can identify areas requiring attention and implement corrective actions when necessary. The availability of organized and accessible information supports more efficient resource allocation and strengthens long-term planning efforts.

Moreover, the adoption of data-driven decision-making contributes to greater accountability and transparency within property management operations. Decisions can be justified through documented financial information and performance indicators, reducing uncertainty and increasing confidence in management actions. As a result, data-driven decision-making serves as a valuable foundation for improving operational efficiency, financial sustainability, and overall property management effectiveness.

Financial Monitoring. The continuous process of observing, recording, and evaluating financial transactions to maintain an accurate and current view of a business's financial condition. It helps organizations track income, expenses, cash flow, and overall financial performance in real time. Effective financial monitoring enables managers to identify potential issues early, make informed decisions, and ensure that financial resources are being used efficiently to achieve business objectives.

Furthermore, financial monitoring plays an important role in maintaining the stability and sustainability of rental property operations by ensuring that financial information remains organized, accurate, and consistently updated. Through regular monitoring of rental income, expenses, outstanding balances, and other financial activities, property owners can evaluate the financial condition of their properties and identify irregularities or potential financial concerns at an early stage. Effective financial monitoring also supports transparency, improves financial control, and enables more efficient management of available resources, contributing to improved operational and strategic decision-making.

Financial Reporting. The process of compiling and presenting financial information in a structured format, such as income statements, expense summaries, and cash flow reports, to provide a comprehensive view of financial performance over a specific period.It enables stakeholders, including managers, investors, and regulatory authorities, to assess the organization's financial health and operational efficiency. Accurate and timely financial reporting supports informed decision-making, ensures transparency, and helps maintain compliance with financial regulations and reporting standards.

Furthermore, financial reporting serves as a critical communication tool that transforms financial data into meaningful and understandable information for decision-makers. By organizing financial transactions into structured reports, stakeholders can evaluate income trends, expenditure patterns, profitability, and overall financial performance over a defined reporting period. This organized presentation of information allows users to gain a clearer understanding of the financial condition of an organization or business operation.

In addition, financial reporting supports accountability by providing documented evidence of financial activities and resource utilization. Accurate reports enable property owners and managers to review financial performance, compare actual results with planned budgets, and identify areas requiring improvement. Regular reporting also helps ensure that important financial information is readily available for review, reducing the likelihood of oversight and improving financial control.

Moreover, financial reporting contributes to strategic planning and long-term decision-making by providing historical and current financial information that can be analyzed to identify trends and patterns. Through the use of financial reports, decision-makers can assess operational efficiency, evaluate the impact of management decisions, and determine appropriate actions for future growth and sustainability. As a result, financial reporting serves as an essential component of effective financial management and supports informed, data-driven decision-making.

Within the context of rental property management, financial reporting enables property owners to monitor rental income, track operational expenses, review outstanding balances, and evaluate the financial performance of individual properties or units. These reports provide valuable insights that assist in budgeting, financial forecasting, and the overall management of rental property operations, making financial reporting a key function of systems such as PropSight.

Booking System. A module within PropSight that allows tenants to submit requests to reserve or occupy available rental units and allows administrators to manage, approve, or decline those requests.It streamlines the reservation process by providing a centralized platform for handling booking applications and tracking their status. The system also helps reduce manual processing, improve communication between tenants and administrators, and ensure that unit availability is updated accurately and efficiently.

Furthermore, the booking system serves as an important mechanism for managing the allocation of available rental units in an organized and systematic manner. By centralizing booking requests within a single platform, the system allows administrators to review applications more efficiently, verify unit availability, and maintain accurate records of tenant reservations. This reduces the likelihood of scheduling conflicts, duplicate bookings, and other issues commonly associated with manual reservation processes.

In addition, the booking system improves the overall experience of both tenants and property administrators by providing a transparent process for submitting, reviewing, and tracking booking requests. Tenants can monitor the status of their applications, while administrators can access relevant booking information needed for decision-making and property management activities. The availability of organized booking records also supports more effective occupancy management and contributes to improved utilization of rental units.

Moreover, the integration of the booking system with other modules of PropSight enables booking information to be linked with tenant records, unit information, and financial transactions. This integration helps maintain data consistency across the system and supports a more comprehensive approach to rental property management. As a result, the booking system contributes not only to operational efficiency but also to improved record management and service delivery within rental property operations.

Maintenance Request. A formal submission by a tenant reporting a problem with their unit or the property and requesting repairs or attention from the property administrator. This request serves as an official record of maintenance-related concerns, enabling property managers to track, prioritize, and address issues in a timely manner. Maintenance requests may include problems such as plumbing leaks, electrical malfunctions, damaged fixtures, structural defects, or other conditions that affect the safety, functionality, or comfort of the property. Proper documentation and management of maintenance requests help improve response times, maintain property quality, and enhance tenant satisfaction by ensuring that reported issues are resolved efficiently.

Adding to this, maintenance requests contribute to the effective management and preservation of rental properties by providing a structured process for identifying, documenting, and resolving property-related issues. Through proper recording and monitoring of maintenance concerns, property administrators can track the status of reported problems, assign priorities based on urgency, and ensure that corrective actions are completed within a reasonable timeframe. The availability of organized maintenance records also assists in evaluating recurring issues, planning preventive maintenance activities, and supporting the long-term upkeep of rental properties.

Tenant Payment Tracking. The recording and monitoring of rental payments made by tenants, including the status of each payment (paid, unpaid, or pending), the date of payment, and the amount covered.It helps property managers maintain accurate financial records and ensures that all rental transactions are properly documented. By providing real-time updates on payment statuses, the system enables timely follow-ups on overdue accounts and supports effective financial management.

What is more, payment tracking enables property owners and administrators to maintain greater control over rental income by providing a clear and organized record of all tenant payment activities. Through systematic monitoring of payment histories, due dates, and outstanding balances, users can quickly determine which accounts are current and which require follow-up actions. This helps reduce the likelihood of overlooked payments and supports more consistent revenue collection.

At the same time, payment tracking contributes to improved financial transparency and accountability by ensuring that all rental transactions are properly recorded and easily accessible for review. Accurate payment records allow property owners to verify transactions, resolve payment-related disputes, and maintain reliable financial documentation. The availability of historical payment information also supports financial analysis and assists in evaluating tenant payment behavior over time.

Also, the integration of payment tracking with other system modules, such as financial reporting and dashboard analytics, enhances the overall effectiveness of rental property management. By automatically updating payment statuses and reflecting changes in financial records, the system provides users with timely and accurate information that supports informed decision-making, efficient account management, and improved oversight of rental property finances.

Dashboard Analytics. A visual summary screen within PropSight that displays charts, graphs, and key financial indicators to help the property owner understand financial performance trends at a glance. It consolidates important data into an easy-to-read format, enabling users to monitor income, expenses, occupancy rates, and other critical metrics efficiently. By providing real-time insights and trend analysis, the dashboard supports informed decision-making and enhances overall property management effectiveness.

In addition, the analytics dashboard serves as a centralized interface for presenting complex financial and operational data in a simplified and visually understandable manner. Rather than requiring users to review large volumes of raw transaction records, the dashboard organizes information into graphical representations and summary indicators that allow important trends and performance measures to be identified more quickly. This improves the accessibility of information and enables users to gain a comprehensive overview of property performance within a single screen.

Building on this, the analytics dashboard enhances monitoring capabilities by providing immediate access to current and historical data related to rental operations. Users can compare financial performance across different reporting periods, evaluate income and expense trends, monitor outstanding balances, and assess the overall financial condition of their rental properties. The availability of these insights supports more proactive management by allowing potential issues and opportunities to be identified at an earlier stage.

On top of this, the integration of dashboard analytics with other modules within PropSight ensures that displayed information remains accurate, current, and consistent with the system's underlying financial records. As new transactions, payments, expenses, or maintenance activities are recorded, relevant dashboard indicators are updated accordingly. This continuous flow of information supports effective financial oversight, improves operational awareness, and strengthens the ability of property owners to make informed, evidence-based management decisions.

Financial Transparency. refers to the availability, accessibility, accuracy, and clarity of financial information related to the operation of rental properties. In the context of this study, financial transparency enables property owners to have a complete and organized view of their rental income, operational expenses, maintenance costs, tenant payment records, and overall financial performance. Through the use of PropSight, financial data is systematically recorded, monitored, and presented in a manner that is easy to understand and interpret. This allows property owners to verify transactions, monitor cash flow, identify financial trends, and evaluate the profitability of their properties with greater confidence. Financial transparency also promotes accountability and reduces the likelihood of errors, omissions, or inconsistencies in financial records by ensuring that all relevant financial information is available in a centralized and reliable system.

Expense Categorization. refers to the systematic process of classifying and organizing property-related expenditures into specific categories based on their nature and purpose. In the context of this study, expense categorization involves grouping financial transactions such as maintenance costs, repair expenses, utility payments, administrative expenses, and other operational costs into clearly defined categories within the PropSight system. This process helps property owners maintain organized financial records, monitor spending patterns, and gain a clearer understanding of how financial resources are being allocated. By categorizing expenses, property owners can more easily identify areas of high expenditure, control unnecessary costs, prepare accurate financial reports, and perform meaningful financial analyses. Furthermore, expense categorization supports data-driven decision-making by providing structured financial information that can be used to evaluate operational efficiency and improve overall property management practices.

Property Owner Empowerment. refers to the enhancement of a property owner's ability to effectively manage, control, and make informed decisions regarding the financial and operational aspects of their rental properties. In the context of this study, property owner empowerment is achieved through the provision of accurate, organized, and timely financial information that enables property owners to improve their understanding of the financial condition of their rental businesses. Through the features of PropSight, property owners are given access to financial monitoring tools, automated reporting functions, and data visualization capabilities that transform raw financial data into meaningful insights. This empowers property owners to move beyond informal and reactive management practices by making decisions based on reliable evidence and real-time information. As a result, they can improve cash flow management, monitor profitability, plan for future expenses, reduce financial uncertainty, and respond more effectively to operational challenges. Ultimately, property owner empowerment contributes to greater confidence, efficiency, and sustainability in managing rental property businesses as a source of income and livelihood.

Transaction History. refers to the complete and chronological record of all financial transactions associated with the operation of a rental property. In the context of this study, transaction history includes records of rental income, tenant payments, outstanding balances, maintenance expenses, utility costs, repair expenditures, and other financial activities processed through the PropSight system. It serves as a centralized source of financial information that allows property owners to review past transactions, verify financial records, monitor cash flow, and track changes in financial performance over time. By maintaining an accurate and organized transaction history, property owners can improve financial transparency, support report generation, facilitate auditing and reconciliation processes, and make more informed decisions regarding the management and sustainability of their rental property businesses.

Financial Insights. refer to the meaningful information, patterns, trends, and interpretations derived from the analysis of financial data related to rental property operations. In the context of this study, financial insights are generated through the monitoring and analysis of rental income, operational expenses, maintenance costs, tenant payment records, and other financial transactions recorded in the PropSight system. These insights provide property owners with a clearer understanding of their financial performance, cash flow status, profitability, and spending patterns. By transforming raw financial data into understandable and actionable information, financial insights help property owners identify opportunities for improvement, detect potential financial issues, evaluate business performance, and make informed, data-driven decisions that support the efficient management and long-term sustainability of their rental property businesses.

Agile Software Development Methodology. An iterative and incremental approach to software development that emphasizes flexibility, continuous improvement, and responsiveness to user feedback, used as the development framework for PropSight.

Entity Relationship Diagram (ERD). A graphical representation of the data entities in a system and the relationships between them, documenting how data objects such as users, tenants, properties, units, payments, and expenses are related in the database.

Data Flow Diagram (DFD). A diagram showing how data moves through a system, depicting the flow of information between external actors, system processes, and data stores.

Three-Tier Architecture. A software design pattern separating a system into three distinct layers: the Presentation Layer (user interface), the Application Layer (business logic), and the Data Layer (database), used in PropSight to promote modularity and maintainability.

User Acceptance Testing (UAT). A testing process in which representative end users evaluate a completed system to confirm that it meets their requirements and is ready for actual use.

MySQL. An open-source relational database management system used by PropSight to store, organize, and manage all financial and operational data.

PHP. A server-side scripting language used to build the backend logic of PropSight, connecting the user interface to the database and processing all system operations.

Management Information System (MIS). A computer-based system that provides managers with organized information to support decision-making, coordination, control, analysis, and visualization of information in an organization. It collects, processes, stores, and distributes data from various operational activities, transforming it into meaningful information that can be used for managerial purposes. In the context of PropSight, the MIS integrates financial records, payment tracking, booking information, and analytical reports into a centralized platform. This enables property owners and administrators to monitor operations efficiently, evaluate performance, and make informed decisions based on accurate and timely data. By improving access to relevant information, an MIS enhances organizational productivity, operational control, and strategic planning.

Cash Flow. The movement of money into and out of a business over a specified period. In the context of rental property management, cash flow refers to the difference between rental income received and operational expenses incurred during the same period.

Outstanding Balance. The amount of money owed by a tenant that has not yet been paid, representing the difference between the total rental fees due and the total payments received for a given period. It reflects any unpaid obligations that remain outstanding after partial or missed payments. Monitoring outstanding balances is essential for maintaining accurate financial records and ensuring the timely collection of rental income. In PropSight, outstanding balances are tracked automatically, allowing property owners and administrators to identify overdue accounts, monitor payment compliance, and take appropriate actions when necessary. This helps improve cash flow management and supports more effective financial oversight of rental properties.

Role-Based Access Control (RBAC). A security mechanism that restricts system access based on the roles assigned to individual users, ensuring that each user can only perform actions and access data appropriate to their designated role.

Purposive Sampling. A non-probability sampling technique in which participants are selected based on specific criteria relevant to the research objectives, used in this study to identify rental property owners who are most likely to provide relevant data about the problem PropSight is designed to address. This sampling method allows researchers to focus on individuals who possess direct experience and knowledge of property management and financial monitoring processes. By selecting participants with relevant expertise, the study can gather more meaningful and detailed insights regarding existing practices, operational difficulties, and system requirements. As a result, purposive sampling helps ensure that the collected data effectively supports the development and evaluation of the proposed system.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Introduction

This chapter presents a review of related foreign and local studies and related literature that provide the theoretical and empirical foundation for PropSight. The review covers studies on digital property management systems, financial monitoring platforms, web-based rental management applications, and the use of data analytics in property and small-business financial management.It examines existing technologies and methodologies that have been implemented to improve operational efficiency, financial transparency, and decision-making processes in rental property management. The chapter also explores the strengths, limitations, and key features of related systems to identify gaps that the proposed system aims to address. Furthermore, it discusses relevant concepts, theories, and frameworks that support the development of an integrated property management and financial monitoring solution. By analyzing previous research and literature, the study establishes the significance of adopting digital tools for managing rental properties and financial records. The insights gathered from these sources serve as a basis for the design, functionality, and implementation of PropSight. Ultimately, the review helps justify the need for the proposed system and demonstrates its potential contribution to improving property management practices. In addition, this chapter highlights the increasing global shift toward digital transformation in property management and financial systems. Recent studies indicate that traditional manual and semi-digital approaches are no longer sufficient to meet the growing complexity of rental operations, particularly in environments where multiple tenants, varying payment schedules, and recurring operational expenses must be managed simultaneously. As a result, researchers have increasingly focused on developing centralized and automated systems that enhance accuracy, efficiency, and accessibility of financial information.

What is more, the literature reveals a strong emphasis on the integration of real-time monitoring and automated reporting features within modern property management systems. These features are widely recognized as essential in reducing delays in financial tracking, minimizing human error, and improving the reliability of financial data used for decision-making. This trend supports the relevance of systems like PropSight, which aim to provide real-time financial visibility and structured reporting capabilities.

The reviewed studies also demonstrate a growing use of data-driven approaches and analytical tools in financial management. By transforming raw financial data into meaningful insights through visualization and reporting mechanisms, property owners are better equipped to identify trends, evaluate performance, and make informed strategic decisions. This aligns with the core objectives of PropSight, which emphasizes structured data presentation and analytical support for rental property management.

At the same time, the literature identifies usability and accessibility as critical success factors in the adoption of property management systems. Many existing platforms are either too complex or not tailored for small to medium-scale property owners, resulting in limited adoption despite their technical capabilities. This gap highlights the need for simplified, user-friendly systems that balance functionality with ease of use, ensuring that non-technical users can effectively manage their financial and operational data.

Finally, this chapter establishes a clear foundation for understanding how digital systems can address long-standing inefficiencies in rental property management. By synthesizing findings from various studies and literature, it provides a comprehensive background that supports the development of PropSight as a practical, scalable, and data-driven solution for modern property management challenges.

2.2 Related Literature

Okeke, Bakare, and Achumie (2024), writing in the International Journal of Management and Entrepreneurship Research, conducted a comprehensive case review of data-driven financial management systems in small and medium enterprises. Their review found that such systems consistently increased financial transparency and gave business owners a competitive advantage by enabling faster and improved decision-making. The study identified the main barriers to adoption among small businesses as limited technical expertise, high initial implementation costs, and the lack of solutions tailored to specific needs. These findings are directly applicable to PropSight, which is designed specifically to address these barriers for small-scale property owners.

Ayankoya, Omotoso, and Ogunlana (2025), writing in the International Journal of Management and Organizational Research, examined the role of data-driven financial optimization in improving the efficiency and resilience of small and medium enterprises. Their study highlighted that the digitalization of financial operations had accelerated significantly, driven by the proliferation of affordable software tools and cloud-based platforms. The authors recommended that SMEs invest in integrated financial management platforms that combine data collection, processing, and reporting in a single system, an approach that aligns closely with PropSight's design.

Mirza and Pokharel (2023), in a systematic literature review examining the impact of digital transformation on the real estate industry, analyzed journals, reports, and case studies from 2000 to 2023. The review found that digital tools, including information systems, data analytics platforms, and cloud-based management solutions, were reshaping business models and transforming how property stakeholders manage and report on financial performance. The authors emphasized that the most impactful implementations combined financial data management, reporting automation, and user-friendly interfaces, an approach that directly informs PropSight's integrated design.

Chiwhane, Bagane, Sourabh, Nagrale, Jha, and Pandey (2023), writing in the International Journal of Intelligent Systems and Applications in Engineering, presented EstimaRent, a data-driven rental housing optimization system designed to help real estate professionals make a more efficient pricing and management decisions. The system used data analysis tools to monitor rental market data, identify pricing trends, assess tenant demand, and evaluate property performance metrics. The core finding that data analytics tools significantly improve the quality of financial decision-making in the rental property context is directly applicable to PropSight.

Waja, Shah, and Nanavati (2021), writing in the International Journal of Engineering Applied Sciences and Technology, reviewed the application of agile software development in modern IT projects. The study highlighted that agile methodologies are particularly well-suited to projects where user requirements are not fully defined at the outset and are expected to evolve through testing and feedback. The authors found that agile practices consistently produced systems better aligned with end-user needs compared to traditional waterfall approaches, supporting PropSight's adoption of Agile methodology.

The broader literature on financial information systems and digital decision-support tools provides additional theoretical and empirical context for PropSight's design and expected impact.

One particularly relevant body of literature concerns the relationship between financial information system adoption and business performance outcomes in small enterprises. Multiple studies reviewed across the period from 2021 to 2025 have found consistent positive associations between the use of organized digital financial management systems and improvements in profitability, cash flow management, and strategic planning quality among small business owners. Ayankoya et al. (2025) synthesized several of these studies in their review and concluded that the availability of real-time, organized financial data is one of the most significant contributors to improved decision-making quality in SME contexts, as it reduces reliance on intuition and gut feeling in favor of evidence-based analysis.

The literature on dashboard design and data visualization also provides important guidance for the development of PropSight's analytics module. Research reviewed by Naeem et al. (2023) and by the broader literature on decision-support systems consistently finds that the effectiveness of financial dashboards depends on several key design factors: the appropriate selection of key performance indicators (KPIs) that are genuinely relevant and actionable for the user, the use of visual formats such as charts and color coding that allow users to quickly identify patterns and anomalies, the minimization of information overload through the careful prioritization of what is displayed on the main dashboard versus accessible through detailed reports, and the responsiveness of the dashboard to real-time data updates so that the information displayed always reflects the current state of the business.

These design principles are directly applied in PropSight's dashboard design. The selection of dashboard KPIs for PropSight, including total monthly income, number and total amount of outstanding balances, monthly expenses, net cash position, and occupancy rate, was guided by both the literature on small-business financial management and by the findings of the user interviews conducted during the planning phase of development. The use of color-coded indicators for payment statuses and trend charts for income and expenses reflects the data visualization best practices identified in the reviewed literature.

The literature on database design for financial information systems also informs PropSight's data architecture. Research on relational database design for small-business financial management consistently emphasizes the importance of data normalization, referential integrity enforcement, and audit trail mechanisms in maintaining the accuracy and reliability of financial records over time. PropSight's MySQL database schema is designed with these principles in mind, using normalized tables, foreign key constraints, and activity logging to ensure that financial records remain consistent, traceable, and reliable even as the volume of transactions grows.

The property technology, or proptech, sector has emerged as one of the most dynamic areas of technology innovation globally over the past decade, with particular acceleration since 2020. Proptech broadly refers to the application of digital technologies to the real estate and property management sectors, encompassing a wide range of tools from online property listing platforms and virtual viewing technologies to artificial intelligence-driven valuation systems and blockchain-based property transaction platforms.

Within the proptech landscape, property management systems (PMS) represent the category most directly relevant to PropSight. Property management systems are software platforms designed to automate and streamline the administrative, financial, and operational functions of property management. At their most basic, they provide digital replacements for the manual record-keeping processes used by property owners and managers: tenant databases, payment ledgers, maintenance logs, and financial reports. At their most advanced, they integrate with smart building sensors, payment gateways, digital leasing platforms, and portfolio analytics tools to provide a comprehensive, real-time view of every aspect of property performance.

The global PMS market is dominated by a relatively small number of large platforms, including Yardi, AppFolio, Buildium, and RealPage, which are designed primarily for the North American market and cater primarily to mid-size to large property management companies. These platforms offer comprehensive functionality but come with price points and implementation complexity that put them well beyond the reach of small-scale landlords in developing markets like the Philippines. A second tier of lower-cost, cloud-based PMS platforms, including Rentec Direct, TenantCloud, and Innago, offer more accessible pricing but are still designed primarily for the US and UK markets and do not fully account for the specific financial management practices, regulatory environment, or technology infrastructure of the Philippine rental market.

The absence of locally developed, affordable, and specifically tailored property management solutions for the Philippine market creates an opportunity for research and development in this area. PropSight is designed to fill part of this gap by providing a focused, accessible financial monitoring tool that addresses the most pressing financial management needs of small-scale landlords in the Philippine context, using technology that is familiar and affordable for its target users.

The broader proptech literature also highlights several important trends that inform the design of PropSight, even though they represent future development directions rather than current features. The integration of mobile payment technologies, including GCash and PayMaya which are widely used in the Philippines, into property management systems is an increasingly important feature in the local market. A future version of PropSight could integrate with these payment platforms to allow tenants to pay rent digitally, with payments automatically recorded in the system and reflected in real-time payment status updates. The incorporation of predictive analytics into property management systems is another significant trend, with advanced platforms now offering

AI-driven forecasting of rental income, occupancy rates, and maintenance costs. While these features are beyond the scope of the current study, the data architecture of PropSight is designed to be compatible with such future enhancements.

The proptech literature also provides important guidance on the usability requirements for property management systems targeting non-technical users. Research consistently finds that the adoption of proptech tools among small-scale property owners is heavily influenced by ease of use, with complex or unintuitive interfaces cited as a major barrier to adoption even when the underlying functionality is relevant and useful. This finding reinforces the emphasis on user-centered design in PropSight's development, and supports the decision to include usability as one of the five primary evaluation criteria for the system.

The relationship between information systems adoption and business performance improvement in small enterprises has been studied extensively across multiple sectors and geographic contexts. A consistent finding across this body of literature is that the introduction of even basic information systems, when well designed and appropriately tailored to the user's context, can produce disproportionately large improvements in operational efficiency and financial visibility. This finding is particularly relevant to the context of PropSight, where the baseline condition, manual record-keeping using notebooks and basic spreadsheets, represents one of the lowest possible points on the technology adoption curve. The potential for improvement from this baseline is correspondingly large.

The concept of financial visibility deserves particular attention in the context of small-scale rental property management. Financial visibility refers to the degree to which a business owner has access to accurate, timely, and organized information about the financial condition of their business. In large organizations, financial visibility is maintained through dedicated accounting teams, enterprise resource planning systems, and regular management reporting processes. In small enterprises, and particularly in micro-enterprises like small-scale rental property businesses, financial visibility is often severely limited by the absence of these resources and structures. The consequences of low financial visibility include delayed recognition of financial problems, reactive rather than proactive management, and a reduced ability to plan and invest strategically.

PropSight addresses the financial visibility problem directly by providing a centralized, organized platform for recording and reviewing all financial transactions associated with a rental property business. The analytics dashboard, in particular, is designed to provide immediate, at-a-glance financial visibility without requiring the property owner to compile or analyze data themselves. By reducing the effort required to maintain financial visibility to the simple act of logging in to the system and reviewing the dashboard, PropSight aims to make data-driven management a realistic and sustainable practice for property owners who currently lack the time, expertise, or tools to maintain adequate financial oversight.

The literature on technology adoption in developing economies provides important context for understanding the challenges and opportunities associated with introducing a system like PropSight into the Philippine rental property market. Research consistently identifies several factors that shape the likelihood of technology adoption among small business operators in developing country contexts: the perceived relevance of the technology to their specific operational challenges; the perceived ease of use and the availability of support for learning to use the technology; the cost of adoption relative to the perceived benefits; and the social and cultural norms around technology use within the business community.

Each of these factors has been considered in the design of PropSight. Perceived relevance is addressed by grounding the system's design in the specific challenges documented in the local literature and in user interviews with actual rental property owners, ensuring that every feature of the system addresses a real and recognized need. Perceived ease of use is addressed through the emphasis on intuitive interface design, real-time validation feedback, and a consistent navigation structure that reduces the learning curve for new users. Cost is addressed by building the system using freely available open-source technologies that do not require expensive licenses, and by designing it to run on basic web hosting infrastructure rather than requiring specialized or costly server environments. Social and cultural norms are addressed by designing the system in a language and format that is appropriate for the Filipino small business context.

The role of trust in technology adoption is also relevant to PropSight's design and potential impact. Research on technology adoption in financial management contexts consistently finds that trust is a critical factor: users must trust that the system will accurately represent their financial data, that their data will be kept secure, and that the system will behave reliably and predictably. Distrust, whether based on past negative experiences with technology or on unfamiliarity with digital systems, is one of the most significant barriers to adoption among potential users who might otherwise benefit from a system like PropSight.

PropSight addresses the trust dimension through several specific design decisions. The security architecture is designed to protect financial data from unauthorized access, reducing the risk that a security breach will erode user confidence in the system. The reliability testing procedures described in Section 3.8 are designed to verify that the system behaves consistently and predictably under a range of operating conditions, building the foundation for user confidence through demonstrated performance. The transparency of the system's financial calculations, which can be verified by the user against their own records during the initial adoption period, further supports trust-building by allowing users to independently confirm that the system is recording and calculating their financial data correctly.

The long-term sustainability of PropSight's impact on the financial management practices of its users is an important consideration that goes beyond the scope of this initial study but deserves attention in the discussion of the system's significance. Research on information system adoption in small business contexts consistently finds that initial adoption, while necessary, is not sufficient to produce lasting changes in management practice. For a system like PropSight to deliver its intended benefits, users must not only adopt the system initially but continue to use it consistently over time, updating records regularly and relying on its reports and dashboard for ongoing financial decision-making.

Sustained adoption requires that the system continue to meet the user's needs as those needs evolve over time. As property owners become more familiar with PropSight and more comfortable with data-driven management, their expectations of the system are likely to increase. Features that seem adequate in the initial adoption phase may become insufficient as users develop more sophisticated financial management practices. This dynamic underscores the importance of building PropSight on a flexible, extensible architecture that can accommodate future enhancements, and of designing the development process to remain responsive to user feedback beyond the initial evaluation phase.

The academic literature on information systems success models provides useful frameworks for evaluating whether a system like PropSight is likely to achieve its intended long-term impact. The DeLone and McLean Information System Success Model, originally proposed in 1992 and updated in 2003, identifies three primary dimensions of information system quality: system quality (the technical quality of the system itself), information quality (the quality of the information the system produces), and service quality (the quality of support available to users). These three dimensions of quality are predicted to influence user satisfaction and system use, which in turn produce individual and organizational net benefits.

Applying this framework to PropSight, system quality corresponds to the reliability, efficiency, security, and usability of the system as assessed in the evaluation phase of this study. Information quality corresponds to the accuracy, completeness, timeliness, and relevance of the financial reports, dashboard indicators, and payment status information that PropSight produces. Service quality, in the context of a student-developed system, corresponds to the availability of documentation, user guides, and responsive developer support during the initial deployment period. All three dimensions are addressed in PropSight's design and evaluation, though the service quality dimension is acknowledged as the most challenging given the resource constraints of a student development project.

Financial management theory provides a rich conceptual backdrop for understanding why systems like PropSight are needed and how they create value for small business operators. At its most fundamental level, financial management theory holds that the purpose of financial information is to support decision-making: decisions about how to allocate resources, how to manage risk, how to price products and services, and how to plan for the future. When financial information is unavailable, incomplete, or inaccurate, the quality of these decisions inevitably suffers.

For rental property owners, the most critical financial management decisions fall into four broad categories. The first is pricing decisions, specifically the determination of appropriate rental rates for different units. Pricing decisions have a direct and immediate impact on revenue, and getting them wrong in either direction has significant consequences: rates that are too low leave money on the table and may not cover the full cost of managing the property, while rates that are too high lead to vacancy and lost income. Good pricing decisions require accurate knowledge of both the costs associated with each unit and the market rates for comparable units in the area.

The second category is investment decisions: decisions about whether to acquire additional properties, invest in improvements to existing properties, or dispose of underperforming assets. These decisions require accurate knowledge of the current and projected financial performance of existing properties, including income trends, expense trends, vacancy rates, and cash flow patterns. Without organized financial records, property owners cannot make these assessments reliably and tend to base investment decisions on intuition and optimism rather than evidence.

The third category is operational decisions: day-to-day decisions about how to manage the property, including maintenance priorities, tenant selection, lease renewal negotiations, and expense management. Operational decisions are made most effectively when the decision-maker has a clear view of the financial implications of different choices. For example, a property owner who can see from the expense summary that maintenance costs for a particular unit have been consistently high over the past twelve months is in a much better position to make an informed decision about whether to invest in a more comprehensive renovation or to increase the rent to cover the elevated maintenance costs.

The fourth category is risk management decisions: decisions about how to protect the property business against financial risks such as tenant default, unexpected large repair costs, and market downturns. Effective risk management requires both awareness of the risks and the financial capacity to absorb or mitigate them. A property owner who maintains an adequate cash reserve for unexpected expenses is better positioned to handle a sudden large repair cost without disrupting their cash flow, and a property owner who monitors tenant payment patterns closely is better positioned to identify and address default risks before they escalate.

PropSight supports all four categories of financial management decisions through its various modules and reports. The income and expense tracking modules provide the data needed for pricing and investment decisions. The cash flow report provides the data needed for risk management decisions, specifically the management of cash reserves and the identification of periods of cash flow stress. The outstanding balance report and real-time payment monitoring support operational decisions about tenant management and debt recovery. The analytics dashboard provides an integrated view of financial performance that supports all four categories simultaneously.

The financial management literature also provides important guidance on the design of financial reports for non-expert users. Research on financial literacy among small business owners consistently finds that many small business operators struggle to interpret traditional financial statements such as income statements and balance sheets, which are designed for use by financially trained professionals. This finding has important implications for the design of PropSight's reporting module, which must present financial information in a format that is meaningful and useful for property owners who may not have formal financial training.

PropSight addresses the financial literacy challenge through the use of simplified report formats that present key financial information in plain language, with clear headings and summary indicators that allow users to understand the main message of the report without needing to read through detailed line-item data. The analytics dashboard, in particular, is designed to communicate the most important financial indicators through visual formats such as charts, graphs, and color-coded status indicators, which research has shown to be more accessible to non-expert users than tabular numerical data. These design choices reflect the principle, articulated in the financial literacy literature, that financial tools create the most value when they are designed to meet users where they are rather than requiring users to develop new technical skills before they can benefit from the tool.

The connection between financial management practice and business sustainability is another important theoretical consideration. Research on small business failure consistently identifies poor financial management, including inadequate record-keeping, insufficient cash flow monitoring, and reactive rather than proactive responses to financial problems, as one of the leading causes of business failure among small enterprises. Rental property businesses are not immune to this risk: a property owner who fails to maintain adequate financial records, monitor payment compliance, and manage expenses effectively is at risk of finding themselves in financial difficulty without adequate warning or the information needed to respond effectively.

By improving the financial management practices of small-scale rental property owners, PropSight contributes to the sustainability of these businesses as sources of income and housing. A property owner who uses PropSight consistently and acts on the financial insights the system provides is better positioned to identify problems early, respond proactively, and maintain the financial health of their rental business over the long term. This sustainability dimension of PropSight's impact is not directly measurable in the evaluation conducted as part of this study, which focuses on short-term usability and functionality rather than long-term business outcomes, but it is the ultimate justification for the system's development and provides the most compelling argument for its continued adoption and development.

2.3.1 Local Studies (Philippines Context)

Magno, Pelayo, and Soberano (2024), writing in the International Journal of Multidisciplinary Research and Analysis, developed an Apartment Rental Management System (ARMS) for a private property in Mandaue City, Cebu. The study identified that the property owner was experiencing significant difficulties managing tenant records, computing billing amounts, monitoring payments, and generating reports using manual methods. The ARMS they developed automated these core processes, recording transactions in real time and consolidating all data in a centralized system. User testing showed that the system improved record accuracy, reduced the time needed for routine administrative tasks, and gave the owner a clearer view of occupancy and payment statuses. This study is the closest existing precedent for PropSight in the Philippine context, and its findings validate the premise that a computerized system can meaningfully improve the operational efficiency of small-scale rental management.

Nercua et al. (2023), writing in the International Journal of Industrial Management, conducted a phenomenological qualitative study on the constraints faced by small-scale landlords in Davao City, Philippines. Through interviews with five landlords, the study identified financial challenges as a consistently significant barrier, with participants reporting difficulty tracking income against expenditures, managing outstanding balances, and making informed investment decisions due to the lack of organized financial records. The study recommended that small-scale landlords adopt structured systems for financial monitoring to improve decision-making quality and profitability. These findings provide strong contextual justification for PropSight, as they document challenges faced by property owners in a setting highly similar to the target users of this system.

Santos et al. (2021), cited in Magno et al. (2024), proposed an IoT-based monitoring system for utility meters in residential rental properties in the Philippines, aimed at improving transparency in utility usage and reducing disputes between landlords and tenants. While this study focuses on utility monitoring rather than comprehensive financial management, it confirms the broader research interest in using digital systems to improve transparency and accountability in the Philippine rental market, and highlights the potential for PropSight's financial monitoring functions to reduce similar disputes in payment records and outstanding balances.

The local research landscape on property management and financial monitoring systems in the Philippines has grown meaningfully in recent years, reflecting the broader national push toward digital transformation across industries. Beyond the three primary local studies reviewed in section 2.3.1, several additional local research contributions and national policy developments provide important context for understanding the environment in which PropSight is being developed.

The Philippine government has made digital transformation a national priority through several policy frameworks, including the Philippine Digital Economy Roadmap and the policies of the Department of Information and Communications Technology (DICT), which have encouraged the adoption of digital tools across sectors including small business and property management. These national policy directions create a favorable environment for the adoption of systems like PropSight, as property owners are increasingly aware of and interested in digital management tools, even if they lack access to suitable, affordable options.

Research on small business financial management in the Philippine context also provides relevant background. Studies on the financial management practices of micro, small, and medium enterprises (MSMEs) in the Philippines consistently find that informal and manual financial tracking is the norm among the smallest operators, and that the adoption of digital financial management tools is associated with improved business performance outcomes. The Department of Trade and Industry (DTI) has noted in its annual MSME development reports that financial management capability is one of the most commonly cited areas where small business owners need support, suggesting a broader market demand for accessible financial management tools beyond the specific context of rental property management.

The local literature on IT capstone projects in Philippine universities also provides useful context. A review of recent capstone research from Philippine information technology programs shows a growing number of studies focused on developing management information systems for specific local small-business contexts, including restaurant management systems, cooperative financial management systems, and various types of inventory and point-of-sale systems. PropSight contributes to this body of work by focusing on the underserved context of rental property financial management, a domain that has received relatively little attention in Philippine IT research compared to retail and food service management. The growing adoption of

Management Information Systems (MIS) among Philippine organizations also provides relevant context for the development of PropSight. Local studies consistently report that computerized information systems improve organizational efficiency by centralizing data, reducing manual workloads, and providing timely access to operational information. These systems allow managers to monitor activities more effectively and make decisions based on accurate and up-to-date records. The principles underlying these systems closely align with the objectives of PropSight, which seeks to provide property owners with reliable financial information that supports data-driven management decisions.

Local research on financial information systems likewise emphasizes the importance of accurate financial records in supporting organizational sustainability. Philippine studies involving small businesses, cooperatives, and service-oriented enterprises have found that digital financial monitoring tools contribute to improved budgeting, expense tracking, revenue monitoring, and financial reporting. Researchers have noted that many small organizations struggle with fragmented financial records and delayed reporting when relying solely on manual processes. The implementation of computerized financial systems has been shown to reduce these challenges by improving record accuracy and providing managers with clearer visibility into financial performance.

Studies involving accounting and financial reporting systems developed in Philippine academic institutions further support the relevance of PropSight. Numerous capstone projects and applied research studies have focused on automating financial transactions, generating financial statements, and improving the accessibility of financial information for decision makers. Common findings indicate that automated reporting features reduce the time required to prepare financial reports while increasing the reliability of financial data. These benefits are particularly important in rental property management, where property owners must regularly monitor rental income, maintenance expenses, utility charges, and tenant payment histories.

Another area of local research relevant to PropSight involves decision-support systems and business analytics applications. As organizations increasingly recognize the value of data in strategic planning, researchers have explored how information systems can transform operational data into actionable insights. Such systems allow managers to identify trends, evaluate performance, and make informed decisions based on historical and current information. The concept of data-driven management reflected in these studies directly supports the objectives of PropSight, which aims to help property owners analyze financial information and make improved decisions regarding property operations and investments.

The Philippine rental housing sector has also experienced gradual modernization due to increasing urbanization and demand for rental accommodations. As rental properties become an increasingly important source of income for many individuals and families, the need for effective management practices becomes more significant. Property owners must balance tenant management responsibilities with financial oversight, often without access to specialized tools designed for their needs. This situation creates a practical opportunity for systems such as PropSight, which combine financial monitoring and reporting capabilities within a platform specifically intended for rental property management.

Furthermore, local studies on digital transformation have highlighted the importance of adopting technology to remain competitive and responsive in a rapidly changing environment. Researchers have observed that organizations that embrace digital solutions are generally better positioned to improve operational efficiency, maintain accurate records, and respond to emerging challenges. Within the context of rental property management, digital transformation can help property owners transition from reactive decision-making based on incomplete information to proactive management supported by comprehensive financial data and reporting tools.

Overall, the body of local literature demonstrates a strong and growing interest in the application of information systems to address operational, financial, and managerial challenges. While relatively few Philippine studies have focused specifically on financial monitoring and reporting systems for rental properties, existing research on management information systems, financial information systems, and rental management solutions collectively supports the development of PropSight. These studies establish a clear need for a system that integrates financial monitoring, reporting, and data-driven decision-making functions into a single platform tailored to the needs of rental property owners. In addition, the increasing adoption of cloud-based and web-enabled systems in local Philippine research highlights a gradual shift toward more flexible and accessible property management solutions. These systems allow property owners to access financial and operational data remotely, improving convenience and responsiveness in managing rental properties, especially in urban settings where mobility and real-time access are important.

Furthermore, several local studies emphasize the persistent gap between available digital solutions and their actual adoption among small-scale property owners. Despite the presence of management systems, many users still rely on manual or semi-digital methods due to limited awareness, technical skills, or access to appropriate systems. This gap highlights the need for more intuitive and user-centered platforms that align with the capabilities of non-technical users, reinforcing the relevance of PropSight's simplified design approach.

Another important observation from local literature is the emphasis on affordability and practicality in system adoption. Many small property owners operate under limited budgets, making costly or complex systems less viable. As a result, research supports the development of lightweight, cost-effective, and efficient systems that focus on essential financial monitoring and reporting features rather than overly complex enterprise-level functionalities.

Moreover, local studies on system usability and user experience consistently highlight that ease of navigation and clarity of information presentation significantly influence user satisfaction and system effectiveness. Systems that are overly complicated tend to discourage continuous use, while those with clear interfaces and straightforward workflows are more likely to be adopted successfully in real-world environments.

Lastly, the integration of real-time data processing and automated reporting in local information systems is increasingly recognized as a key factor in improving financial decision-making. By reducing delays in data availability and minimizing manual intervention, these systems enhance the reliability and timeliness of financial insights, further supporting the development and relevance of PropSight in addressing the needs of Philippine rental property management.

2.3.2 Foreign Studies

Buradkar, Kori, Ruikar, and Galfat (2022) developed a Property Rental Management System aimed at replacing the manual, paper-based rental management processes common among property managers. Published in the International Journal of Computer Science and Mobile Computing, the study identified that manual record-keeping frequently resulted in delays in updating tenant records and made it difficult to track rental payment histories accurately. Their system used a centralized web-based database that allowed property managers to maintain property listings, access tenant information, and log rental transactions more organized. The study concluded that transitioning to a centralized digital platform produced clear improvements in operational efficiency and data accuracy. This research directly supports the core argument of PropSight that a centralized digital system is substantially more effective than manual methods for managing rental property records.

Kadam, Mhatre, and Potenavaru (2024), writing in IJARCCE, presented an online property management system addressing the need for an automated, user-friendly platform for booking, managing, and monitoring rental properties. Their system allowed users to search for available properties, create accounts, book units online, and manage listings through an admin interface with reporting and analytics. The study highlighted the importance of a clear user interface and centralized data management in improving both administrative efficiency and the tenant experience. The design principles and functional scope of their system closely parallel those of PropSight, particularly the admin dashboard, booking management, and reporting features.

Dai (2020), in a study published in Ingenierie des Systemes d'Information, introduced a Property Management Information System (PMIS) based on data management strategies informed by modern information technologies. The study noted that traditional property management methods were no longer adequate to handle the growing volume and complexity of real estate transactions. The PMIS developed by Dai enabled property managers to store, process, and retrieve large amounts of property-related data with greater speed and precision, improving both operational efficiency and financial decision-making quality. Dai's conceptual framework informed the design of PropSight's three-tier data architecture and its emphasis on reliable, structured data management.

Ghimire and Charters (2022), writing in the journal Software published by MDPI, examined the impact of agile development practices on software project outcomes. Their empirical study found that agile practices, including iterative sprint-based development, continuous feedback integration, and collaborative team structures, were positively associated with imrpoved project quality, higher team satisfaction, and more responsive adaptation to changing requirements. These findings reinforce the rationale for PropSight's adoption of Agile methodology.

Naeem, Ahmed Rana, and Nasir (2023), in a systematic review published in Smart Construction and Sustainable Cities by Springer Nature, examined the technologies and tools driving digital transformation in real estate. Through thematic analysis of peer-reviewed publications, the study identified four key themes: information communication technologies, data collection technologies, data networking tools, and digital decision-making systems. The review concluded that real-time monitoring and data-driven decision support were among the most impactful technology contributions to property management efficiency, and noted that small operators in developing economies remained behind in adoption, a gap that PropSight directly aims to address.

Beyond the foreign studies reviewed in section 2.1, a broader review of the international literature reveals a consistent pattern of research activity directed at improving the efficiency and transparency of property management through digital systems.

A particularly notable trend in the international literature from 2022 to 2025 is the growing integration of cloud computing and software-as-a-service (SaaS) models into property management platforms. According to the IMARC Group's market analysis (2024), cloud-based deployments accounted for the largest share of the global property management software market in 2023 and are projected to continue growing, driven by the flexibility, scalability, and remote access advantages they offer over on-premises installations. While PropSight is designed initially as a locally hosted web application using XAMPP, the system's architecture is compatible with cloud deployment, and the use of Supabase as a potential database layer provides a pathway to cloud-based operation in future versions.

Another trend is the increasing use of data analytics and business intelligence tools in property management software to go beyond basic record-keeping and provide property managers with genuinely predictive and strategic financial insights. Research reviewed by Agarwal et al. (2023), cited in Kadam et al. (2024), found that the integration of machine learning and AI into property management systems was already enabling dynamic pricing, predictive maintenance scheduling, and tenant behavior analysis in more advanced platforms. While these capabilities are outside the scope of PropSight in its current version, they represent a natural direction for future development, and the data architecture of PropSight is designed with future expansion in mind.

The international literature also includes a significant body of work on the usability challenges of property management software, particularly for users without formal technical training. Studies from the United Kingdom, Australia, and Malaysia have all noted that property management system adoption rates among small-scale landlords remain lower than expected, with usability barriers, including complex interfaces, steep learning curves, and the need for technical support, cited as major obstacles. These findings directly informed the user-centered design approach adopted for PropSight, with particular emphasis on simplifying the interface, minimizing the number of steps required to complete common tasks, and providing immediate, helpful feedback when users make errors.

The trend toward mobile-first property management solutions is also worth noting in the context of the international literature. The IMARC Group (2024) identified the increasing adoption of mobile-first solutions as a significant driver of property management software market growth, particularly in emerging markets where smartphone penetration is high but desktop computer access is more limited. While PropSight is currently designed as a desktop-first web application with responsive capabilities, for future development it is recommended to consider a dedicated mobile interface or native mobile application as a priority enhancement, given the reality that many small-scale landlords in the Philippines manage their properties using smartphones as their primary digital device. Ma (2021), writing in Applied Mathematics and Nonlinear Sciences, developed a residential Property Management Information System based on browser/server architecture to improve the efficiency of property administration. The study found that traditional property management methods often resulted in fragmented records, delayed information retrieval, and inefficient data processing. The proposed system centralized property, tenant, and transaction data within a unified platform, resulting in improved operational efficiency and more reliable information management. These findings support PropSight's emphasis on centralized financial and rental data management.

Chandra and Hermeindito (2024), writing in Enrichment: Journal of Management, examined the factors influencing user adoption of a property management dashboard using the Technology Acceptance Model (TAM). Their findings indicated that perceived usefulness, ease of use, and trust significantly influenced user acceptance of digital property management systems. The study reinforces the importance of usability and intuitive interface design, principles that are incorporated into PropSight's dashboard and reporting features.

Valks, Arkesteijn, Koutamanis, and Den Heijer (2021), in a study published in Buildings, explored the role of dashboards in supporting data-driven decision-making. The researchers found that visual dashboards improved the interpretation of operational data and enabled managers to identify trends and performance indicators more effectively. These findings are particularly relevant to PropSight's financial reporting dashboard, which seeks to transform raw financial data into actionable insights for property owners.

Cheng (2021), through a study conducted at the University of Hong Kong, investigated the application of Property Technology (PropTech) solutions in rental housing management. The research concluded that digital technologies significantly improve operational transparency, information accessibility, and management efficiency. The study further emphasized that data-driven systems are becoming increasingly important in modern property management environments, supporting the conceptual foundation of PropSight.

Amadi-Echendu (2023), writing in the South African Journal of Information Management, examined the use of integrated information systems in managing property-related processes. The study found that organizations using unified digital platforms achieved improved information visibility, improved coordination, and more effective decision-making compared to organizations relying on fragmented systems. These findings align closely with PropSight's objective of integrating financial monitoring, reporting, and rental management functions into a single platform. Recent developments in property technology (PropTech) literature further emphasize the shift toward intelligent and automated property management ecosystems that integrate financial monitoring, tenant management, and predictive analytics into a single platform. This evolution highlights a growing demand for systems that not only store and process data but also actively support decision-making through real-time insights and automated reporting functionalities, reinforcing the relevance of PropSight's design direction.

In addition, several studies emphasize the increasing importance of interoperability and system integration in modern property management platforms. Contemporary systems are being designed to connect with external financial tools, payment gateways, and cloud-based databases to ensure seamless data flow and reduce redundancy. This trend supports the conceptual basis of PropSight, which prioritizes centralized data handling while maintaining the potential for future integration with broader digital ecosystems.

Furthermore, the literature consistently highlights scalability as a key requirement in property management systems, particularly for small to medium-scale property owners whose operations may expand over time. Systems that are scalable allow users to manage increasing numbers of tenants, properties, and transactions without experiencing performance degradation or data management difficulties. This reinforces the importance of PropSight's structured database design and modular system architecture.

Another emerging focus in foreign studies is the use of automated validation and error-checking mechanisms within financial management systems. These mechanisms are designed to detect inconsistencies, duplicate entries, and missing information in real time, thereby improving data integrity and reducing the reliance on manual verification.

Such features are increasingly recognized as essential in ensuring the reliability of financial records in digital property management environments.

Lastly, the international literature also points to the growing significance of user accessibility and digital inclusivity in system adoption. Studies emphasize that systems designed for non-technical users must prioritize simplicity, clarity, and minimal learning curves to ensure widespread usability. This reinforces the importance of user-centered design principles in PropSight, ensuring that it remains accessible to small-scale property owners who may not have advanced technical backgrounds.

2.4 Synthesis of the Study

The reviewed foreign studies, local studies, and related literature collectively point to several important and consistent conclusions that justify and guide the development of PropSight.

The most fundamental conclusion is that digital, technology-driven management systems represent a significant and well-evidenced improvement over manual approaches to property management. Studies by Buradkar et al. (2022), Kadam et al. (2024), Magno et al. (2024), and Dai (2020) have all demonstrated that transitioning from manual methods to centralized digital platforms produces measurable gains in operational efficiency, data accuracy, record accessibility, and administrative responsiveness.

The reviewed local studies are especially valuable because they document the specific challenges faced by property owners in the Philippine setting. Nercua et al.

(2023) confirmed that financial management difficulties are among the most significant operational constraints facing small-scale landlords in Philippine cities. Magno et al. (2024) demonstrated that a computerized system can substantially reduce these difficulties. These findings place PropSight in direct response to a documented, locally relevant need.

An important distinction also emerges from the review between systems focused on operational management and those that provide financial monitoring and analytical reporting. While existing local systems have automated day-to-day operations effectively, they have not placed equal emphasis on financial analysis, report generation, and data-driven decision support. The studies by Okeke et al. (2024), Ayankoya et al. (2025), Chiwhane et al. (2023), and Mirza and Pokharel (2023) all highlight the transformative potential of data-driven financial tools, yet note that these remain underused among small businesses and property managers in developing economies.

The proptech literature reviewed in Section 2.3 further confirms that the global direction of property management technology is toward integrated, data-driven platforms that combine operational management with financial monitoring and analytical decision support. PropSight's design positions it at the intersection of these trends, making it a timely and relevant contribution to both the local research landscape and the broader proptech field.

Based on this synthesis, PropSight fills a specific and documented gap by combining the operational management features validated in existing rental management system research with the financial monitoring, automated reporting, and analytics tools highlighted in the financial management and digital transformation literature. By targeting small to medium-scale property owners in the Philippine context with an accessible, purpose-built web-based platform, PropSight directly addresses the needs identified in local studies while building on design principles and technological approaches validated internationally. The reviewed studies also reveal a strong consensus regarding the importance of centralized information management in modern property administration. Both local and foreign researchers consistently emphasize that fragmented records and manual documentation processes contribute to inefficiencies, data inconsistencies, and difficulties in generating timely reports. The adoption of centralized databases and integrated management systems has been shown to improve information accessibility, reduce duplication of records, and support more accurate monitoring of operational and financial activities. These findings support PropSight's use of a centralized database architecture designed to consolidate rental, tenant, payment, and financial information within a single platform.

Another significant theme emerging from the reviewed literature is the growing importance of data quality in supporting managerial decision-making. The effectiveness of financial monitoring and reporting systems depends heavily on the accuracy, completeness, and consistency of the data being collected and processed. Several studies noted that organizations relying on manual methods often encounter challenges related to missing information, delayed updates, and reporting inaccuracies. By automating data recording and storage processes, systems such as PropSight can help ensure that property owners have access to more reliable information when evaluating financial performance and making operational decisions.

The literature further highlights the increasing role of reporting dashboards and visual analytics in management information systems. Modern users require more than simple data storage capabilities; they increasingly expect systems to present information in ways that facilitate interpretation and action. Dashboards, charts, and automated reports have been identified as effective tools for communicating key performance indicators and financial trends. The integration of reporting and visualization features within PropSight reflects these findings and contributes to its goal of supporting data-driven property management practices.

A review of both local and international studies also demonstrates the importance of usability and accessibility in determining the success of information systems. Regardless of technical sophistication, systems are unlikely to achieve their intended benefits if users find them difficult to learn or operate. Research consistently indicates that user-friendly interfaces, streamlined workflows, and clear navigation structures contribute significantly to user acceptance and long-term system adoption. These considerations are particularly relevant to PropSight, which is intended for small and medium-scale property owners who may have limited experience with specialized management software.

Another observation derived from the literature is the increasing convergence of property management and financial management technologies. Traditionally, property management systems focused primarily on tenant records, lease administration, and occupancy monitoring, while financial functions were often handled separately through accounting software or manual processes. Recent developments indicate a growing preference for integrated platforms capable of managing both operational and financial activities within a unified environment. This trend reinforces the relevance of PropSight's approach, which combines rental property management functions with financial monitoring and reporting capabilities to provide a more comprehensive solution.

Adding to this, the reviewed literature suggests that digital transformation is no longer limited to large organizations with substantial technological resources. Advances in web technologies, cloud computing, and database management systems have made sophisticated information systems increasingly accessible to small businesses and independent operators. As a result, small-scale landlords and property owners are now better positioned to adopt digital tools that were previously available only to larger property management firms. PropSight aligns with this development by providing an affordable and accessible platform specifically designed for the needs of smaller property management operations.

On the whole, the collective evidence presented in the reviewed studies and literature provides strong support for the development of PropSight. The recurring themes of digital transformation, centralized information management, financial monitoring, automated reporting, data-driven decision-making, usability, and system integration consistently appear across both local and international research. These findings not only validate the need for the proposed system but also provide a clear foundation for its design, functionality, and intended contribution to improving rental property management practices within the Philippine context.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This study uses a developmental research design, the most appropriate methodology for studies in information technology education that aim to create, test, and evaluate new software systems intended to solve practical, real-world problems. Developmental research, also known as design and development research, provides a systematic framework for both the creation of technological artifacts and the evaluation of their effectiveness in actual or near-actual use contexts.

The developmental research process in this study begins with a thorough analysis of the current financial management practices of small-scale rental property owners, drawing on the reviewed literature and primary data gathered through user interviews. The findings of this needs analysis directly inform the system's functional requirements, design decisions, and evaluation criteria. Following the design and development phases, the completed system is evaluated by a defined group of users and evaluators.

The choice of developmental research is supported by the nature of the problem PropSight is designed to address. The research problem, that small-scale landlords lack an accessible, purpose-built digital tool for financial monitoring and reporting, is a practical one that calls for a practical solution. Developmental research is the appropriate methodology for such problems because it centers on creating and validating that solution rather than on purely descriptive or explanatory inquiry.

What is more, developmental research emphasizes the iterative improvement of a system throughout its development lifecycle. Rather than focusing solely on the final product, this approach allows researchers to continuously refine system features, interface components, and functional requirements based on identified user needs and evaluation results. This ensures that the developed system remains aligned with the expectations and operational requirements of its intended users.

At the same time, the developmental research design supports the integration of both theoretical and practical perspectives during system creation. Concepts and findings derived from related literature, existing technologies, and previous studies are translated into functional system components that address real-world challenges. This connection between research and implementation strengthens the relevance and applicability of the developed solution.

The methodology also provides a structured process for assessing whether the proposed system effectively addresses the identified problems. Through systematic evaluation procedures, researchers can determine the extent to which the system improves financial monitoring, reporting efficiency, data organization, and overall property management practices. The evaluation results provide evidence regarding the effectiveness, usability, and reliability of the developed system.

Also, developmental research is particularly suitable for information technology projects because it accommodates the design, construction, testing, and validation of software-based solutions within a single research framework. This enables the researchers to examine not only the technical functionality of the system but also its practical value to end users in real-world operational settings.

Ultimately, the use of developmental research design ensures that PropSight is developed through a systematic, evidence-based process that combines problem analysis, system development, and user evaluation. This approach helps ensure that the resulting system is both technically sound and capable of addressing the financial management needs of small to medium-scale rental property owners.

3.2 System Development Methodology

The development of PropSight is guided by the Agile Software Development Methodology. Agile is an iterative and incremental approach to software development that emphasizes collaboration, flexibility, continuous testing, and the progressive refinement of features based on feedback gathered at each stage of the process. Rather than requiring a complete and fixed specification of all requirements before development begins, Agile allows the team to plan, build, test, and review in repeated cycles, with each cycle adding new functionality and refining existing features.

The choice of Agile for PropSight is based on several practical considerations. First, the system's requirements are expected to evolve through user testing and feedback. Second, Agile's emphasis on iterative testing means that defects and design issues can be identified and corrected early. Third, the collaborative structure of Agile supports efficient division of work among the five researchers while maintaining overall project coherence.

Ghimire and Charters (2022) found that agile teams consistently reported improved project outcomes in terms of quality, responsiveness to change, and alignment with user needs compared to teams using plan-driven methods. Waja, Shah, and Nanavati (2021) similarly found that agile methodologies were especially effective for projects where requirements were not fully defined at the start, a condition that applies directly to PropSight.

In the context of PropSight, the Agile methodology enables the development team to gradually implement and refine core system functionalities such as financial transaction recording, rental income tracking, expense monitoring, tenant payment management, financial report generation, and data visualization features. By developing these components incrementally, the researchers can evaluate each feature's effectiveness and usability before proceeding to subsequent development stages, ensuring that the final system aligns closely with user requirements.

In addition, Agile supports continuous testing and validation throughout the development process. Each iteration provides an opportunity to verify the accuracy of financial calculations, reporting functions, and data processing mechanisms while identifying potential issues early in development. This approach reduces the risk of major defects appearing during later stages of implementation and contributes to the overall reliability and quality of the system.

The methodology also facilitates the progressive enhancement of the system's user interface and user experience. Feedback obtained during testing phases can be incorporated into succeeding development cycles, allowing improvements to navigation, dashboard presentation, report accessibility, and overall system usability. This iterative refinement process helps ensure that the platform remains intuitive and accessible for property owners with varying levels of technical expertise.

Building on this, Agile provides flexibility in integrating additional features and improvements that may emerge during development. As user needs become clearer through testing and evaluation, the development team can adjust priorities and implement enhancements without requiring substantial restructuring of the entire project. This adaptability is particularly valuable for information systems such as PropSight, where practical user requirements may evolve as stakeholders interact with working system prototypes.

The Agile development process for PropSight is organized into five phases, each described in detail below:

Phase 1: Planning

During the planning phase, the research team identifies and documents the functional and non-functional requirements of PropSight through a combination of literature review findings and direct interviews with potential end users. The team reviews existing research on rental management systems and financial monitoring tools, identifies the key features needed by small-scale property owners, and creates a product backlog: a prioritized list of all desired system features and functions. The planning phase also covers the selection of the technology stack, the identification of development roles and responsibilities among team members, and the establishment of a project timeline divided into development sprints. A stakeholder analysis is also conducted during this phase to identify all groups who will interact with or be affected by the system, including property owners, tenants, and system administrators, and to understand the specific needs and expectations of each group. The goal is to ensure that the development team has a clear, shared understanding of what PropSight needs to do before construction begins.

On top of this, the planning phase includes the identification of the system's core modules and functional components. Based on the gathered requirements, the researchers determine the features necessary for effective financial monitoring and rental property management, including income recording, expense tracking, tenant management, payment monitoring, financial report generation, and dashboard-based data visualization. These features are prioritized according to their importance and relevance to the needs of the target users.

During this phase, the researchers also evaluate and select the appropriate development tools and technologies that will be used throughout the project. Considerations such as system performance, ease of development, scalability, security, and compatibility are taken into account when choosing the software, frameworks, and database technologies that will support the implementation of PropSight. This process helps ensure that the selected technology stack can effectively support the system's functional and technical requirements.

Beyond that, preliminary system planning activities include the preparation of initial design documents, workflow diagrams, process models, and database planning structures. These planning artifacts provide a visual representation of how information will flow through the system and how different components will interact with one another. Establishing these foundations early helps minimize development issues and provides clear guidance during the subsequent design and construction phases.

Risk identification and mitigation planning are also considered during this stage. Potential challenges such as requirement changes, development delays, technical difficulties, data management concerns, and testing issues are identified and assessed. By anticipating these risks early, the development team can prepare appropriate strategies to minimize their impact on the overall project timeline and system quality.

Phase 2: System Design

In the system design phase, the research team translates the requirements from planning into a comprehensive set of design artifacts that document the intended structure, behavior, and visual presentation of PropSight. These include the Data Flow Diagram (DFD), which maps how data flows through the system; the Entity Relationship Diagram (ERD), which defines the data entities and their relationships; the Use Case Diagram, which identifies the key interactions between the system and its primary users; and user interface wireframes that guide the visual design of the system's pages and navigation. Database normalization is applied during this phase to ensure that the MySQL schema is organized efficiently and that data redundancy is minimized. The user interface wireframes are developed with reference to the usability principles identified in the literature review, prioritizing clarity, consistency, and ease of navigation for non-technical users.

Along with this, the system design phase establishes the overall architecture of PropSight by defining how the various system modules interact with one another and how information moves between the user interface, application logic, and database components. This architectural planning ensures that the system remains organized, scalable, and capable of supporting future enhancements without requiring significant structural modifications.

During this phase, detailed designs for the system's functional modules are also prepared. These modules include tenant management, property management, payment monitoring, income and expense recording, financial report generation, and dashboard analytics. The relationships and interactions among these modules are carefully defined to ensure that data is processed consistently and displayed accurately throughout the system.

Adding to this, special attention is given to the design of the financial monitoring and reporting components. The researchers identify the specific financial information that must be collected, processed, and presented to users, including rental income, operational expenses, outstanding balances, and payment histories. Appropriate layouts, report formats, and dashboard elements are designed to facilitate the effective presentation of financial information and support data-driven decision-making.

Security and data integrity considerations are likewise incorporated into the system design. Access controls, user authentication mechanisms, and validation procedures are planned to help protect sensitive financial and tenant information from unauthorized access or inaccurate data entry. These design considerations contribute to the reliability and trustworthiness of the system.

The system design phase also includes the preparation of navigation structures and workflow sequences that guide users through various system processes. By carefully organizing menus, forms, reports, and dashboard components, the researchers aim to create a user-friendly environment that enables property owners to perform tasks efficiently while minimizing confusion and unnecessary complexity.

Phase 3: Development

The development phase is where the system is actually built, with the team constructing components across the three architectural layers in line with the design specifications. Development proceeds in a series of sprints, each targeting a specific functional module or set of features. The first sprints focus on building the system's core infrastructure: the database schema, user authentication, and the basic navigation structure of the interface. Later sprints develop the primary functional modules, including property and unit management, tenant and payment tracking, expense recording, financial report generation, and the analytics dashboard. Each sprint ends with an internal review of completed features, and any identified issues are addressed before the next sprint begins. Version control is used throughout to maintain a complete history of all code changes and to facilitate collaborative development among team members.

During this phase, the researchers integrate the various system modules to ensure that information flows accurately and consistently throughout the platform. Data entered through tenant management, property management, payment monitoring, and expense recording modules is automatically processed and stored within the database, allowing the system to generate updated financial information and reports whenever needed. This integration ensures that all components of PropSight function as a unified and centralized system.

What is more, the development phase includes the implementation of the financial monitoring and reporting features that serve as the core functionality of PropSight. These features are designed to automatically record financial transactions, calculate income and expenses, monitor outstanding balances, and generate financial summaries that assist property owners in evaluating the performance of their rental properties. Dashboard components and data visualization elements are also developed during this stage to present financial information in a more organized and understandable format.

The researchers also implement data validation mechanisms to improve the accuracy and reliability of system records. Input validation rules, error handling procedures, and transaction verification processes are incorporated into the system to minimize incorrect entries, incomplete records, and inconsistencies in financial data. These measures help maintain data integrity and support the production of accurate reports and analytics.

At the same time, regular integration testing is conducted throughout development to verify that newly implemented features operate correctly with existing system components. This continuous testing approach enables the development team to identify and resolve technical issues early, reducing the likelihood of significant problems during later stages of the project. Feedback gathered during sprint reviews is likewise used to refine system functionality and improve the overall user experience.

The development phase concludes with the completion of all planned modules and the preparation of the fully functional system for comprehensive testing and evaluation. At this stage, the researchers verify that the implemented features align with the defined system requirements and objectives established during the planning and design phases.

Phase 4: Testing

The testing phase is conducted both during and after development. During development, each completed module is subjected to unit testing to verify that individual functions perform correctly. Once multiple modules are complete, integration testing checks that they work together and that data flows correctly between the frontend, backend, and database layers. After all major modules are done, system testing evaluates the full PropSight system against its functional and non-functional requirements under realistic conditions. Finally, user acceptance testing involves representative end users in assessing the system's practical utility, ease of use, and overall quality. Issues identified during UAT are documented, prioritized, and addressed before the final evaluation.

Also, the testing phase verifies the accuracy and reliability of the system's financial monitoring and reporting functionalities. Test cases are conducted to ensure that rental income, operational expenses, tenant payments, and outstanding balances are correctly recorded, processed, and reflected throughout the system. The generated reports and dashboard summaries are also examined to confirm that calculations, totals, and financial indicators are accurate and consistent with the underlying data.

In addition, usability testing is performed to evaluate how effectively users can interact with the system. The researchers assess the clarity of the user interface, ease of navigation, accessibility of features, and overall user experience. Feedback gathered from users during testing helps identify areas where improvements can be made to enhance system usability and reduce potential difficulties encountered during operation.

The testing phase also includes validation of data integrity and security mechanisms implemented within the system. Input validation controls, authentication procedures, and access restrictions are tested to ensure that unauthorized access is prevented and that invalid or incomplete data cannot compromise the accuracy of system records. These tests contribute to maintaining the reliability and trustworthiness of the information stored within PropSight.

Performance testing is likewise conducted to determine how the system responds under normal operating conditions. The researchers evaluate response times, data retrieval efficiency, report generation speed, and overall system stability while processing typical user transactions. This helps ensure that the system can provide timely access to financial information and maintain acceptable performance during regular use.

Upon completion of all testing activities, the identified issues, defects, and improvement recommendations are reviewed by the development team. Necessary revisions and refinements are then implemented to enhance system functionality, usability, and reliability before the final deployment and evaluation of PropSight.

Phase 5: Evaluation

The evaluation phase is the final stage of the Agile development cycle for PropSight. During this phase, the system is formally assessed by a defined group of evaluators, including potential end users and student researchers, against the five criteria of functionality, usability, reliability, security, and efficiency. Evaluators perform a defined set of tasks within the system and provide structured feedback through survey instruments. The results are analyzed to assess the degree to which PropSight achieves its stated objectives and to identify areas for potential improvement.

Building on this, the evaluation phase determines whether the system effectively addresses the specific problems identified at the beginning of the study. Particular attention is given to assessing the system's ability to improve financial record management, facilitate the generation of accurate reports, provide real-time monitoring of financial activities, and support data-driven decision-making for rental property owners. The evaluation findings help establish whether the implemented features successfully meet the operational and financial management needs of the target users.

On top of this, evaluators assess the quality and usefulness of the system's financial monitoring and reporting capabilities. The generated reports, dashboard visualizations, payment monitoring functions, and expense tracking features are reviewed to determine whether they provide meaningful, accurate, and actionable information. This assessment helps verify that the system delivers practical value in supporting rental property management activities.

The evaluation phase also examines user satisfaction and overall acceptance of the system. Feedback regarding ease of use, interface design, feature accessibility, and system responsiveness is collected and analyzed to determine how effectively users can interact with the platform. Understanding user perceptions is essential in evaluating the likelihood of successful adoption and continued use of the system in real-world settings.

Beyond that, the results of the evaluation serve as the basis for identifying future enhancements and development opportunities. Recommendations provided by evaluators may reveal additional features, interface improvements, reporting capabilities, or performance optimizations that can further improve the effectiveness of PropSight. These insights contribute to the long-term sustainability and continuous improvement of the system.

Ultimately, the evaluation phase provides comprehensive evidence regarding the overall quality, effectiveness, and practical applicability of PropSight. By systematically analyzing evaluator feedback and performance outcomes, the researchers are able to determine the extent to which the system fulfills its intended purpose and contributes to improved financial monitoring and reporting within rental property management.

3.3 System Architecture

PropSight is built on a three-tier architectural model that separates the system into three distinct functional layers: the Presentation Layer, the Application Layer, and the Data Layer. This architecture promotes modularity, maintainability, and scalability by giving each layer a well-defined role that can be updated or expanded without disrupting the others.

The Presentation Layer serves as the front-end component of PropSight and acts as the primary point of interaction between users and the system. This layer is designed to provide a user-friendly, responsive, and accessible interface that enables users to perform various tasks efficiently. Through this layer, users can navigate the system, view property listings and information, submit inquiries, manage their accounts, and access other relevant features offered by the platform. The Presentation Layer focuses on delivering a seamless user experience by presenting information in a clear and organized manner while ensuring that user actions are accurately captured and transmitted to the underlying system components.

Along with this, the separation of the user interface from the system's business logic allows developers to update the visual design, improve usability, or introduce new interface features without affecting the core operations of the application. This separation not only enhances flexibility but also simplifies maintenance and future upgrades. As user expectations and technological trends evolve, the Presentation Layer can be continuously refined to improve user satisfaction while maintaining the stability of the entire system.

The Application Layer, commonly referred to as the business logic layer, serves as the core processing component of PropSight. It acts as an intermediary between the Presentation Layer and the Data Layer, ensuring that all user requests are processed according to the system's predefined rules and operational requirements. Whenever a user performs an action, such as searching for properties, submitting information, or managing records, the Application Layer validates the request, executes the necessary business processes, and determines the appropriate response.

This layer is responsible for implementing the system's functional requirements, enforcing business policies, managing workflows, handling authentication and authorization processes, and coordinating communication among different system modules. By centralizing these operations, the Application Layer ensures consistency, accuracy, and reliability throughout the application. Furthermore, this structured approach allows developers to modify business rules, integrate additional functionalities, or enhance existing processes without affecting the user interface or database architecture. As a result, the system remains adaptable to changing business needs while maintaining high performance and operational efficiency.

The Data Layer forms the foundation of PropSight by managing the storage, organization, retrieval, and security of all system data. This layer is responsible for maintaining essential information such as property records, user profiles, transaction histories, system logs, and other data required for the platform's operations. It provides a structured mechanism for storing information within a database while ensuring that data remains accurate, consistent, and readily accessible whenever requested by the Application Layer.

One of the primary responsibilities of the Data Layer is to safeguard data integrity through effective database management practices, including validation, indexing, backup procedures, and access control mechanisms. By isolating database operations from the application's business logic and user interface, the system achieves greater reliability and security. This separation also allows database administrators and developers to optimize database performance, implement data protection measures, and scale storage resources as the volume of information increases over time. Consequently, the Data Layer supports the long-term growth and sustainability of the PropSight platform while ensuring the efficient handling of large amounts of data.

The implementation of a three-tier architecture provides PropSight with a robust and well-structured foundation that supports both current operational requirements and future system growth. By dividing the system into distinct layers with specific responsibilities, the architecture promotes modularity, making the application easier to develop, maintain, and enhance. Changes made within one layer can typically be implemented without requiring significant modifications to the other layers, reducing development complexity and minimizing the risk of system-wide issues.

What is more, this architectural approach improves system scalability by allowing each layer to be expanded independently based on performance demands and user growth. It also strengthens security by ensuring that users interact only with the Presentation Layer while all data access requests are processed and validated through the Application Layer before reaching the database. This controlled flow of information reduces potential security vulnerabilities and helps protect sensitive data from unauthorized access.

At the same time, the three-tier architecture facilitates teamwork among developers by enabling different development teams to focus on separate layers simultaneously. This improves productivity, accelerates development cycles, and simplifies testing and troubleshooting activities. Through its modular design, enhanced security, maintainability, and scalability, the three-tier architecture enables PropSight to deliver a reliable, efficient, and future-ready property management and information system capable of adapting to evolving technological and business requirements.

3.3.1 Presentation Layer (User Interface)

The Presentation Layer is the part of the system that users directly interact with. It includes all visual elements, forms, navigation menus, dashboards, and data displays that make up the user-facing interface of PropSight.

For the Property Owner (Admin), this includes a financial dashboard showing total monthly income, outstanding balances, and expense summaries; property and unit management screens; a tenant management interface; payment recording and tracking screens; expense entry and categorization forms; financial report generation interfaces; and a maintenance request management screen.

For the tenant (User), the layer provides screens for registering an account, browsing available units, submitting booking requests, checking payment status, and submitting maintenance requests. The Presentation Layer is built using HTML5, CSS3, and JavaScript.

The Presentation Layer serves as the communication bridge between users and the underlying system functions. Its primary responsibility is to present information in a clear, organized, and user-friendly manner while allowing users to efficiently perform system-related tasks. Since the target users of PropSight include small to medium-scale property owners who may possess varying levels of technical expertise, particular emphasis is placed on simplicity, consistency, and ease of use throughout the interface design.

The visual design of the system follows standard user interface design principles to improve accessibility and usability. Consistent navigation structures, clearly labeled menus, organized forms, and intuitive dashboard layouts are incorporated to minimize user confusion and reduce the learning curve associated with system adoption. These design considerations help ensure that users can quickly locate information, complete transactions, and access reports with minimal effort.

The dashboard component of the Presentation Layer plays a particularly important role in supporting data-driven decision-making. Through the use of summary cards, charts, graphs, and visual indicators, financial information is presented in a format that is easier to understand than raw numerical data. This allows property owners to quickly assess the financial condition of their rental properties, monitor payment activities, identify trends, and detect potential issues that may require immediate attention.

Also, the Presentation Layer is designed to provide responsive feedback to user actions. Confirmation messages, notifications, validation prompts, and status updates help users understand the outcome of their actions and reduce the likelihood of errors during system use. This interactive communication between the user and the system contributes to a more efficient and reliable user experience.

In addition, the interface is designed to support future enhancements and additional functionalities. The modular structure of the Presentation Layer allows new screens, forms, reports, and dashboard components to be incorporated as system requirements evolve. This flexibility supports the long-term maintainability and scalability of PropSight while ensuring that the user experience remains consistent across future versions of the system.

The Presentation Layer also contributes significantly to the overall efficiency and effectiveness of PropSight by ensuring that information is delivered to users in a timely and meaningful manner. Rather than requiring users to manually search through large volumes of data, the interface is designed to prioritize relevant information based on user roles and responsibilities. Property owners are provided with immediate access to key financial indicators, occupancy statistics, tenant information, and maintenance activities, while tenants are presented with features and information that directly support their rental experience. This role-based presentation of information helps improve productivity, reduces unnecessary complexity, and enables users to focus on the tasks that are most relevant to them.

To further enhance usability, the Presentation Layer incorporates form validation mechanisms that verify user input before data is submitted to the system. Required fields, input formatting rules, and error notifications help prevent incomplete or inaccurate data entry, thereby improving data quality and reducing the likelihood of processing errors. These validation features contribute to a smoother user experience by providing immediate guidance and reducing the need for corrective actions later in the workflow.

The Presentation Layer also supports effective communication between users and the system through the implementation of dynamic interface elements. Real-time notifications, status indicators, confirmation dialogs, and alert messages provide users with continuous feedback regarding system activities and transaction outcomes. For example, users receive confirmation when payments are successfully recorded, booking requests are submitted, or maintenance requests are processed. Such feedback mechanisms increase user confidence in the system and help ensure that important actions are completed successfully.

Security considerations are likewise integrated into the design of the Presentation Layer. User authentication interfaces, secure login forms, password management features, and session management controls help protect user accounts and sensitive information. Access to specific pages and functionalities is restricted according to user roles and permissions, ensuring that users can only view and interact with information relevant to their authorized responsibilities. These measures strengthen the overall security posture of the system while maintaining a convenient and accessible user experience.

Another important characteristic of the Presentation Layer is its support for responsive web design. The interface is developed to function effectively across a variety of devices, including desktop computers, laptops, tablets, and smartphones. Responsive layouts, flexible content structures, and adaptive design techniques enable users to access system functionalities regardless of screen size or device type. This capability is particularly valuable for property owners and tenants who may need to monitor property activities, review financial information, or submit requests while away from their primary workstations.

From a software engineering perspective, the Presentation Layer is designed with maintainability and extensibility in mind. The use of structured front-end development practices and modular interface components allows developers to efficiently update existing features or introduce new functionality without requiring substantial modifications to the entire system. As PropSight continues to evolve, additional dashboards, reporting modules, communication tools, and user services can be integrated into the interface while preserving a consistent visual appearance and user experience.

All in all, the Presentation Layer serves as more than simply the visual component of PropSight. It functions as a comprehensive interaction environment that facilitates communication, supports decision-making, enhances usability, promotes data accuracy, and ensures secure access to system resources. Through its carefully designed interface and user-centered features, the Presentation Layer plays a critical role in enabling property owners and tenants to effectively utilize the system and achieve their respective objectives.

3.3.2 Application Layer (Backend Logic)

The Application Layer contains all the business logic, data processing functions, and workflow management mechanisms that drive PropSight's core operations. It sits between the Presentation Layer and the Data Layer, receiving inputs from the interface, processing them according to business rules, executing database operations, and returning results to the interface.

Key functions include: validating and processing booking requests; managing financial transaction records; calculating outstanding tenant balances; routing maintenance requests; generating formatted financial reports; and enforcing role-based access controls. The Application Layer is built in PHP, using prepared statements to protect against SQL injection.

The Application Layer serves as the central processing unit of PropSight, ensuring that all system operations are executed in a structured and consistent manner. It is responsible for implementing the business rules that govern how financial data, tenant information, and property records are handled within the system. By separating logic from both the user interface and data storage, this layer enhances modularity and simplifies system maintenance and future enhancements.

In terms of financial management functionality, the Application Layer processes all computations related to rental income, operational expenses, and tenant payment balances. It ensures that all financial calculations are performed accurately and consistently before being stored or displayed to the user. This includes automatically updating outstanding balances, aggregating monthly income and expenses, and preparing data required for financial reporting and dashboard visualization.

The layer also plays a crucial role in enforcing data validation and integrity rules. Before any information is committed to the database, the Application Layer checks for completeness, correctness, and consistency of input data. This reduces the likelihood of errors, duplicate entries, or incomplete records that may affect financial reporting accuracy. It also ensures that only authorized and properly formatted data is processed by the system.

Building on this, the Application Layer manages communication between different system modules, ensuring seamless interaction among tenant management, property management, payment tracking, expense recording, and reporting functions. This integration allows PropSight to operate as a unified system where changes in one module are immediately reflected across related components.

Security mechanisms are also implemented within this layer, particularly in handling authentication and authorization processes. Role-based access control ensures that property owners and tenants can only access features and data relevant to their roles. Additionally, the use of secure coding practices, such as prepared statements in PHP, helps protect the system from common web vulnerabilities, including SQL injection attacks.

In general, the Application Layer is essential in transforming raw user inputs into meaningful financial information and system outputs. It ensures that PropSight operates efficiently, securely, and accurately while supporting the system's goal of providing reliable financial monitoring and reporting for rental property management.

3.3.3 Data Layer (Database)

The Data Layer is responsible for the persistent storage, organization, retrieval, and management of all data in PropSight. It maintains the integrity, consistency, and reliability of all financial and operational records.

The Data Layer stores user account information and role assignments; tenant profiles and contact details; property and unit records; rental payment records by tenant, unit, and date; operational expense records by category and property; maintenance request data; and generated financial report summaries.

The system uses MySQL as its relational database management system, providing robust support for complex relational data structures and data integrity enforcement through foreign key constraints. Supabase is considered as a potential cloud-based deployment option for future versions.

This section presents a detailed explanation of PropSight's system architecture, including the design choices made for each layer, the reasons behind those choices, and the technologies used to support the system's functional and non-functional requirements.

The choice of a three-tier architecture for PropSight is motivated by several well-established software engineering principles. Separation of concerns is the most fundamental of these: by dividing the system into three distinct layers, each with a clearly defined responsibility, the architecture ensures that changes to one layer do not require corresponding changes to the others. For example, if the visual design of the user interface needs to be updated, this can be done by modifying only the Presentation Layer without touching the business logic in the Application Layer or the data schema in the Data Layer. This separation makes the system easier to maintain, test, and extend over time.

Scalability is a second important motivation for the three-tier architecture. If the number of properties or tenants managed through PropSight grows significantly, the Data Layer can be scaled independently, for example by moving to a more powerful database server or migrating to a cloud-based database service, without requiring changes to the Application or Presentation Layers. Similarly, if the Application Layer needs to be updated to support new business logic, this can be done without disrupting the user interface or the database structure.

The Presentation Layer of PropSight is built using a standard web technology stack of HTML, CSS, and JavaScript. HTML provides the semantic markup structure for all system pages, using elements such as forms, tables, navigation menus, and data display containers. CSS is used for all visual styling, including the color scheme, typography, layout, and responsive design breakpoints that allow the interface to adapt to different screen sizes. JavaScript is used for client-side interactivity, including real-time form validation, dynamic content updates without full page reloads, and interactive dashboard chart rendering.

The Application Layer is built in PHP, which serves as the server-side scripting language that processes all system requests and coordinates between the user interface and the database. PHP was selected for the Application Layer for several reasons. First, it is widely supported by low-cost shared hosting environments, which is important for a system designed to be accessible to small-scale property owners with limited IT infrastructure budgets. Second, it has extensive documentation, a large developer community, and a rich ecosystem of libraries and frameworks, making it a practical choice for a student development team. Third, it integrates seamlessly with MySQL, providing efficient and reliable database connectivity. The Application Layer uses PHP's MySQLi extension for database interactions, implementing prepared statements and input validation to enhance security and help protect against SQL injection attacks.

The Data Layer uses MySQL as the primary relational database management system. The database schema is designed following the principles of relational database normalization, with data organized into related tables that minimize redundancy and ensure data integrity. The main tables in the PropSight database correspond directly to the entities identified in the Entity Relationship Diagram: Users, Tenants, Properties, Units, Payments, Expenses, MaintenanceRequests, FinancialReports, and ActivityLogs. Foreign key constraints are used to enforce referential integrity between related tables, ensuring that records in one table always correspond to valid records in the tables they reference.

The PropSight database schema also includes several design features specifically to support the financial monitoring and reporting functions of the system. Payment records include fields for both the payment amount and the expected amount, allowing the system to automatically calculate the payment status and outstanding balance for each tenant. Expense records include a category field that allows expenses to be grouped and summarized by type in financial reports. A financial_period field is included in both payment and expense records to allow the system to generate reports for specific time periods without requiring complex date range queries.

3.4 Tools and Technologies Used

The development of PropSight uses a carefully selected stack of tools and technologies that are widely adopted in web development and well-suited to the system's technical requirements. The table below summarizes the primary tools and technologies used:

Category	Tool / Technology	Role in PropSight
Frontend	HTML	Structural markup for all system pages, forms, and interface screens
Frontend	CSS	Visual styling, responsive layout, and dashboard presentation
Frontend	JavaScript	Client-side interactivity, dynamic content updates, and form validation
Backend	PHP	Server-side business logic, database connectivity, session management, and report generation
Database	MySQL	Relational data storage, query processing, and data integrity enforcement
Database (Cloud)	Railway	Cloud-based PostgreSQL option considered for deployment
IDE	Visual Studio Code	Primary code editing environment with extensions for PHP, HTML, CSS, and JavaScript
Local Server	XAMPP	Local development environment hosting Apache, PHP, and MySQL

3.5 Data Gathering Techniques

The data gathered for this study comes from three primary sources: direct interviews with potential property owner end users, direct observation of their current management practices, and a systematic review of related existing studies. These techniques were chosen to ensure that the system's design is grounded in the actual needs and experiences of its target users.

Semi-structured interviews are conducted with a purposely selected group of small to medium-scale rental property owners who currently manage their properties independently. Participants are selected on the basis that they own and manage at least one rental property with two or more units and do not use dedicated property management software. The interview guide covers the following areas: how the participant currently tracks rental income and expenses; specific challenges they face in managing financial records; what information they find most useful for evaluating property performance; and what features are expected to form a digital management system.

Interviews are conducted in person or via video call, are audio-recorded with participant consent, and are transcribed for analysis. Key themes from the interviews are used to refine the system's functional requirements and prioritize features in the product backlog. Direct observation of how potential users currently manage their financial records is conducted alongside the interviews where possible, supplementing the interview findings with firsthand observation data.

The selection of research participants for this study follows purposive sampling principles, which ensure that the individuals interviewed and observed are those most likely to provide relevant and useful data for the research questions being investigated. Purposive sampling is the appropriate sampling strategy for this study because the research is not attempting to produce statistically representative findings that can be generalized to the entire population of rental property owners in the Philippines. Rather, it is seeking to understand the experiences and needs of a specific subset of that population: small to medium-scale landlords who manage their properties independently using manual or semi-manual financial management methods.

The criteria for participant selection are designed to ensure relevance and diversity. Relevance is ensured by requiring that all participants are currently managing rental properties and are experiencing the financial management challenges that PropSight is designed to address. Diversity is encouraged by seeking participants who vary in the number of units they manage, the type of properties they own, their geographic location within the Iloilo region, and their level of prior experience with digital management tools. This diversity increases the likelihood that the requirements identified through the interviews and observations will reflect the needs of a broad range of potential PropSight users rather than the specific preferences of a narrow and homogeneous group.

The number of participants selected for the interview and observation phase of the study is guided by the concept of information saturation, the point at which additional interviews no longer reveal substantively new themes or requirements. In qualitative research, information saturation is typically reached after a relatively small number of in-depth interviews, often between five and fifteen participants, depending on the homogeneity of the participant group and the complexity of the research questions. For this study, the research team plans to conduct interviews with at least five to eight property owners, with additional interviews conducted as needed until no new functional requirements or design insights are emerging from the data.

The analysis of interview and observation data follows a thematic analysis process adapted from the qualitative research methodology described by Braun and Clarke (2006) and widely used in information systems research. Thematic analysis involves reading and re-reading the interview transcripts and observation notes, identifying recurring ideas and patterns, grouping these into themes, and using the themes to develop a structured understanding of the research problem and its implications for system design. For this study, the thematic analysis focuses specifically on identifying functional requirements (what the system needs to do), usability requirements (how the system needs to behave to be usable by non-technical users), and priority requirements (which features are most urgently needed versus which are secondary or aspirational).

The output of the thematic analysis is a structured requirements document that lists all identified functional and non-functional requirements, organized by priority and mapped to specific system modules. This requirements document forms the basis for the PropSight product backlog, which guides the Agile development process. The requirements document is reviewed and validated by the research team before the start of the first development sprint, and is updated as new requirements are identified during user testing in later phases of the study.

An important methodological consideration in this study is the management of the relationship between the researchers as system developers and the research participants as potential end users. There is a risk that participants may feel pressure to provide positive feedback about the system out of a desire to please the researchers, particularly if the interview and testing sessions are conducted by team members who are also the developers. To mitigate this risk, the research team separates the roles of interviewer and developer where possible, and designs the evaluation questionnaire to include both positively and negatively worded items to reduce acquiescence bias. Participants are also reminded at the beginning of each session that honest, critical feedback is more valuable to the research than positive feedback, and that there are no right or wrong answers.

The ethical dimensions of the research are also carefully considered. All participants provide informed consent before taking part in interviews, observations, or testing sessions. Consent forms explain the purpose of the research, the use of audio recordings, the confidentiality of responses, and the right to withdraw at any time without penalty. No personal financial information of participants or their tenants is collected, shared, or stored as part of the research process. All data collected is stored securely and is accessible only to members of the research team.

To further strengthen the reliability and validity of the gathered data, the study applies triangulation by comparing and cross-verifying information obtained from interviews, direct observations, and literature review findings. This ensures that the identified system requirements are not based solely on subjective user responses but are supported by multiple sources of evidence. Triangulation enhances the credibility of the results and helps reduce bias in interpreting user needs.

In addition to thematic analysis, the study employs a structured coding process during data analysis. Interview transcripts and observation notes are systematically coded by assigning labels to significant statements related to financial tracking, reporting challenges, system expectations, and usability concerns. These codes are then grouped into categories and later refined into overarching themes that directly inform system design decisions and feature prioritization.

The direct observation component of the study follows a non-participant observation approach, where researchers document how property owners currently perform tasks such as recording rental payments, computing expenses, updating tenant records, and generating financial summaries. Particular attention is given to inefficiencies, repetitive processes, and error-prone activities in manual or semi-digital systems. These observations provide contextual understanding that complements the interview findings.

To ensure trustworthiness of the qualitative data, the study follows established criteria such as credibility, dependability, confirmability, and transferability. Credibility is supported through participant validation, where key findings are briefly confirmed with selected respondents. Dependability is ensured through consistent application of the interview guide and observation checklist across all participants. Confirmability is maintained by keeping detailed documentation of raw data, coding decisions, and theme development. Transferability is addressed by clearly defining participant selection criteria and context of the study.

Beyond that, a semi-structured observation checklist is utilized during data gathering to ensure consistency across different participants. This checklist includes indicators such as the method of recording income and expenses, frequency of financial updates, tools used (notebook, spreadsheet, or digital apps), and common difficulties encountered during financial tracking. This structured approach ensures that observation data is systematically captured and aligned with the research objectives.

3.6 Testing Procedures

The testing of PropSight uses a multi-level strategy that verifies the system's correctness, reliability, and usability at progressively higher levels of integration and complexity. Each level addresses a different dimension of system quality and provides feedback that informs further development and refinement.

At the initial level, unit testing is conducted to evaluate individual components of the system. Each function within PropSight, such as data input validation, financial computation, report generation, and database queries, is tested independently to ensure that it produces correct and expected outputs. This level of testing helps identify logical and syntax errors early in the development process before modules are integrated into the larger system.

Following unit testing, integration testing is performed to assess the interaction between different system modules. This includes verifying that data flows correctly between the Presentation Layer, Application Layer, and Data Layer. For example, a recorded rental payment in the user interface must be accurately processed by the application logic and properly stored and retrieved from the database without data loss or inconsistency. This ensures that all components of PropSight function cohesively as a unified system.

System testing is then conducted to evaluate the complete and fully integrated version of PropSight. At this stage, the system is tested against both functional and non-functional requirements, including performance, reliability, usability, and security. Realistic scenarios such as multiple tenant payments, simultaneous expense entries, and financial report generation are simulated to determine whether the system performs effectively under typical operational conditions.

User Acceptance Testing (UAT) serves as the final validation stage, where actual or representative end users evaluate the system based on real-world usage scenarios. Participants perform tasks such as recording payments, generating financial reports, and viewing dashboards to determine whether the system meets their practical needs. Feedback gathered during this stage is used to identify usability issues, missing features, and areas requiring improvement before final deployment.

All test results are systematically documented, including identified defects, test outcomes, and corrective actions taken. This documentation ensures traceability and provides a structured record of system quality assurance activities. The iterative nature of testing allows the development team to continuously refine PropSight until it meets the required standards for functionality, accuracy, and user satisfaction.

3.6.1 Unit Testing

Unit testing is performed on individual system components and functions to verify that each performs its intended task correctly under a range of input conditions. Unit tests are written and executed by the development team during each sprint, enabling early detection and correction of defects before they affect other parts of the system. For PropSight, unit tests cover functions such as payment recording, expense categorization, report calculation, user authentication, and form validation. Each test case specifies the input, the expected output, and the actual output, and results are documented for review. Any function that fails its unit test is corrected and retested before the sprint review.

In addition to functional verification, unit testing in PropSight also includes boundary value testing and invalid input testing to ensure that system components behave correctly under both expected and unexpected conditions. For example, payment input fields are tested with valid amounts, zero values, negative values, and non-numeric inputs to confirm that the system correctly accepts or rejects data based on defined validation rules.

Each unit test is designed with clear test scenarios that align with the system's business logic requirements. This includes verifying that financial computations such as total income, total expenses, and outstanding balances are calculated accurately based on stored transaction records. The correctness of these computations is critical to ensuring the reliability of generated financial reports and dashboard summaries.

Error handling mechanisms are also evaluated during unit testing. The system is tested to ensure that appropriate error messages are displayed when invalid inputs are detected and that such errors do not cause system crashes or data corruption. This helps maintain system stability and improves user experience by providing clear feedback during data entry.

Along with this, unit testing supports code quality and maintainability by encouraging modular development practices. Each function is designed to perform a specific task, making it easier to isolate, test, and debug individual components without affecting the rest of the system. This modular approach also facilitates future updates and enhancements to PropSight.

All unit testing results are logged and reviewed during sprint evaluations. These records include details of passed and failed test cases, identified issues, corrective actions, and retesting outcomes. This documentation ensures transparency in the testing process and provides a clear audit trail of system improvements throughout development.

3.6.2 Integration Testing

Integration testing examines how different modules of the system interact and verifies that data flows correctly across the boundaries between the frontend interface, the PHP backend logic, and the MySQL database. Integration tests check scenarios such as the correct storage of payments entered through the interface, the accurate retrieval and display of financial report data, the proper enforcement of role-based access controls, and the correct routing of maintenance requests from tenant submission to admin review. Integration testing is conducted after each sprint in which new modules are connected to previously developed components.

In addition to verifying data flow between system components, integration testing also evaluates the consistency of business logic execution across modules. This ensures that calculations and operations performed in one module remain accurate when used or referenced by another module. For example, when a tenant payment is recorded, the system must not only store the transaction correctly but also immediately reflect the updated outstanding balance and income summary in the financial dashboard and reporting modules.

The integration testing process also checks the synchronization of real-time updates across different system functions. Any changes made in tenant records, payment entries, or expense logs are verified to ensure that they are consistently reflected in all relevant system outputs without delay or data mismatch. This is essential for maintaining the reliability of PropSight's financial monitoring capabilities.

Adding to this, integration testing includes the validation of system workflows that involve multiple sequential processes. These workflows include booking approval processes, payment recording to report generation pipelines, and maintenance request handling from submission to resolution tracking. Each step in these workflows is tested to ensure smooth transitions between modules without loss or corruption of data.

Error propagation between modules is also evaluated during integration testing. The system is tested to ensure that errors originating from one module are properly handled and do not negatively affect other connected components. Proper exception handling and fallback mechanisms are verified to maintain system stability during unexpected conditions.

Taken together, integration testing ensures that PropSight functions as a unified and interconnected system rather than a collection of independent modules. It confirms that all components work together seamlessly to deliver accurate, consistent, and reliable financial and property management operations.

3.6.3 System Testing

System testing evaluates the complete, fully integrated PropSight system against its functional and non-functional requirements under conditions that closely simulate real-world use. System tests cover all major use cases, including income recording, expense tracking, report generation, payment monitoring, and maintenance request management, as well as performance and reliability under typical load conditions. Failures identified during system testing are documented in a defect tracking log, prioritized by severity, and corrected before the system proceeds to user acceptance testing.

In addition to validating functional requirements, system testing also evaluates key non-functional aspects of PropSight, including performance, usability, security, and reliability. Performance testing is conducted to assess how the system behaves under typical and increased workloads, such as multiple concurrent users recording payments or generating financial reports simultaneously. This ensures that the system maintains acceptable response times and does not experience significant delays during peak usage.

Security testing is also performed to verify that sensitive financial and tenant data is properly protected. This includes testing authentication mechanisms, session handling, and access restrictions to ensure that unauthorized users cannot access restricted features or information. Particular attention is given to role-based access controls to confirm that property owners and tenants can only access functionalities relevant to their assigned roles.

Usability aspects are assessed during system testing to ensure that the interface remains intuitive and easy to navigate even when all modules are integrated. This includes checking consistency in navigation flows, clarity of system messages, and the accessibility of key features such as dashboards, reports, and transaction forms. The goal is to ensure that system complexity does not negatively affect user experience.

Reliability testing is also carried out to determine the system's stability over continuous operation. The system is observed for unexpected crashes, data inconsistencies, or performance degradation during extended use. This ensures that PropSight can operate consistently in real-world deployment scenarios without frequent interruptions.

All identified issues during system testing are recorded in a structured defect log, including details such as the nature of the defect, affected modules, severity level, and resolution status. This documentation supports systematic debugging and ensures that all critical issues are resolved before proceeding to user acceptance testing.

System testing also verifies that all integrated components of PropSight function cohesively within a production-like environment. Unlike unit and integration testing, which focus on individual modules and their interactions, system testing evaluates the behavior of the entire application as a complete solution. This holistic approach ensures that all functional requirements specified during system analysis and design are properly implemented and operate as intended when accessed through actual user workflows.

A significant focus of system testing is the validation of end-to-end business processes. These processes include tenant registration, unit reservation and booking, payment recording and monitoring, expense management, maintenance request submission and resolution, and financial report generation. Each workflow is executed from beginning to end to verify that data is correctly processed at every stage and that users receive accurate outputs and system responses. This comprehensive validation helps ensure that real-world operational scenarios can be completed successfully without interruptions or inconsistencies.

Another critical area assessed during system testing is data accuracy and reporting integrity. Since PropSight serves as a financial monitoring and property management platform, the correctness of financial calculations is essential. Testing is conducted to verify that rental payments, expenses, outstanding balances, and income summaries are accurately reflected across dashboards, reports, and transaction records. Generated financial reports are compared against expected results to confirm that calculations remain consistent and reliable under various operating conditions.

System testing further evaluates the effectiveness of data validation mechanisms implemented throughout the application. User inputs are tested using both valid and invalid data to ensure that the system appropriately accepts correct information while preventing erroneous, incomplete, or unauthorized entries. Validation rules, error messages, and exception handling procedures are assessed to confirm that users receive clear guidance when input errors occur, thereby improving overall data quality and system reliability.

Compatibility testing is also included as part of the system testing process to ensure that PropSight performs consistently across different web browsers, operating systems, and device configurations. The system interface is evaluated on desktop and mobile platforms to verify that layouts, navigation elements, forms, and reports remain functional and visually consistent regardless of the user's environment. This testing helps maximize accessibility and ensures a positive user experience across multiple devices.

What is more, recovery and fault-tolerance testing are conducted to assess the system's ability to handle unexpected events such as network interruptions, server errors, or database connectivity issues. These tests evaluate whether the application can recover gracefully from failures without causing data loss or compromising system integrity. Proper recovery mechanisms are essential for maintaining business continuity and protecting critical financial and property records.

System testing also examines auditability and record-tracking capabilities within PropSight. Administrative actions, payment transactions, maintenance updates, and report generation activities are reviewed to ensure that appropriate records are maintained for monitoring and accountability purposes. These capabilities support transparency and facilitate future troubleshooting, auditing, and management activities.

At the same time, the testing process assesses the scalability of the system by evaluating its ability to accommodate increasing numbers of users, properties, tenants, and financial transactions. Simulated growth scenarios are used to identify potential performance bottlenecks and determine whether the system architecture can effectively support future expansion. This assessment helps ensure that PropSight remains responsive and reliable as organizational requirements evolve over time.

Upon completion of system testing, a comprehensive evaluation report is prepared summarizing all test cases executed, results obtained, identified defects, corrective actions implemented, and overall system readiness. Only after all critical and high-priority issues have been successfully resolved is the system considered ready to proceed to User Acceptance Testing (UAT). Through this rigorous testing process, PropSight is validated as a dependable, secure, efficient, and fully functional property management and financial monitoring solution capable of meeting the operational needs of property owners and tenants alike.

3.6.4 User Acceptance Testing (UAT)

User acceptance testing is the final stage of the testing process. It involves representative end users, including small-scale property owners and student evaluator groups, in assessing the system's practical usefulness, ease of use, and overall quality. Participants are asked to complete a defined set of representative tasks within PropSight, including recording a rental payment, entering an expense, generating a monthly income report, and submitting a maintenance request. Evaluators observe their interactions and note usability issues or missing features. After completing the tasks, participants provide structured feedback through the evaluation questionnaire.

In addition to task completion, user acceptance testing also evaluates the overall satisfaction of end users with the system based on predefined evaluation criteria such as functionality, usability, reliability, efficiency, and security. These criteria ensure that the system is assessed not only on whether it works, but also on how well it supports real-world usage scenarios in rental property management.

During the testing session, participants' interactions with the system are systematically observed to identify difficulties such as navigation confusion, input errors, or delays in completing tasks. These observations are recorded alongside participant feedback to provide a comprehensive understanding of both user experience and system performance.

The evaluation questionnaire utilizes a structured rating scale to quantify user perceptions of PropSight. Responses are analyzed to determine the overall acceptability of the system and to identify areas that require improvement. This quantitative feedback is complemented by qualitative comments provided by participants, which offer deeper insights into user expectations and potential enhancements.

After data collection, all feedback from user acceptance testing is compiled, categorized, and reviewed by the development team. Critical issues and recurring concerns are prioritized for resolution before final deployment of the system. This iterative feedback process ensures that PropSight aligns closely with the needs and expectations of its intended users.

3.7 Evaluation Criteria

The evaluation framework for PropSight is grounded in established models for assessing the quality of information systems, adapted to the specific context and objectives of this study.

The primary evaluation model used in this study is the ISO/IEC 25010 Systems and Software Quality Requirements and Evaluation (SQuaRE) standard, which provides a comprehensive framework for assessing software quality across multiple dimensions. From the eight quality characteristics defined in ISO/IEC 25010, five were selected as the most relevant to PropSight's objectives and target user population: functional suitability (corresponding to the functionality criterion in this study), usability, reliability, security, and performance efficiency (corresponding to the efficiency criterion in this study).

The functional suitability characteristic in ISO/IEC 25010 encompasses three sub-characteristics: functional completeness, which asks whether the system provides all the functions needed by users for their specified tasks; functional correctness, which asks whether the system provides the right results with the needed degree of precision; and functional appropriateness, which asks whether the functions facilitate the accomplishment of specified tasks and objectives. All three sub-characteristics are assessed in PropSight's evaluation through the user acceptance testing process, in which evaluators attempt to complete all major system tasks and rate the completeness, correctness, and appropriateness of the results.

The usability characteristic in ISO/IEC 25010 encompasses sub-characteristics including learnability, operability, user error protection, and user interface aesthetics. These dimensions are assessed in PropSight's evaluation through a combination of direct observation during user acceptance testing and structured questionnaire responses that ask evaluators to rate the ease of learning to use the system, the ease of completing tasks without errors, and the overall quality of the interface design.

The evaluation questionnaire administered during user acceptance testing is adapted from the System Usability Scale (SUS), a widely used and validated instrument for assessing software usability developed by Brooke (1996). The SUS consists of ten items rated on a five-point Likert scale, covering dimensions of usability including learnability, efficiency, and overall satisfaction. The SUS score is calculated on a scale of 0 to 100, with scores above 70 generally considered acceptable, scores above 80 considered good, and scores above 90 considered excellent. In addition to the SUS items, the PropSight evaluation questionnaire includes additional items specific to the five evaluation criteria of this study: functionality, reliability, security, and efficiency.

The evaluation results will be analyzed using descriptive statistics, including means, standard deviations, and frequency distributions for each evaluation item and criterion. These statistics will be used to assess the overall quality of PropSight and to identify specific areas of strength and areas requiring further development.

3.7.1 Functionality

Functionality measures whether PropSight delivers all the features specified in the system requirements and performs them correctly under normal operating conditions. Evaluators verify that all major use cases can be completed without errors or system failures.

Functionality also assesses the completeness and correctness of system outputs in relation to user inputs and system processes. This includes verifying that financial computations such as income totals, expense summaries, and outstanding balances are calculated accurately and consistently across different modules of PropSight. Evaluators also check whether system features such as report generation, payment tracking, and tenant management behave according to the defined system requirements without missing or incomplete outputs.

Also, functionality testing ensures that all system modules are properly integrated and accessible through the user interface. This means that each feature must not only exist but must also be reachable, operational, and responsive within the intended workflow. Any discrepancies between expected system behavior and actual performance are documented as functional issues for correction prior to final deployment.

3.7.2 Usability

Usability measures how easy it is for target users to learn to use PropSight, navigate its interface, and complete their intended tasks without significant difficulty. Key dimensions include the clarity of the interface design, the consistency of navigation patterns, the helpfulness of feedback messages, and overall user satisfaction.

Usability also evaluates the system’s learnability, particularly how quickly first-time users can understand and operate the core features of PropSight without extensive training. This is important for small to medium-scale property owners who may not have advanced technical backgrounds. The system is assessed based on how intuitively users can perform essential tasks such as recording payments, generating reports, and navigating between modules.

Another important aspect of usability is task efficiency, which measures how effectively users can complete operations with minimal steps and time. PropSight is evaluated on whether its interface design reduces unnecessary actions and streamlines common workflows such as financial entry, report generation, and tenant management.

Error prevention and recovery are also considered in usability evaluation. The system is assessed based on how well it prevents user mistakes through input validation and how clearly it guides users when errors occur. This includes evaluating whether error messages are understandable and whether users can easily correct mistakes without system disruption.

In addition, usability testing considers interface consistency and accessibility across different devices and screen sizes. This ensures that users experience a uniform interface structure and interaction flow, contributing to a more predictable and comfortable user experience.

3.7.3 Reliability

Reliability measures the consistency and dependability of PropSight in performing its functions correctly and without failure. A reliable system produces consistent results for identical inputs, handles errors gracefully, and maintains the integrity of financial records across all transactions.

Reliability also evaluates the system's ability to maintain stable performance during continuous or repeated use over time. This includes assessing whether PropSight can operate for extended periods without system crashes, unexpected shutdowns, or degradation in performance. The system is expected to consistently support core operations such as payment recording, report generation, and data retrieval without interruption.

Building on this, reliability considers the system's fault tolerance and its ability to handle unexpected conditions without losing or corrupting data. For example, if an error occurs during data entry or processing, the system is intended to prevent partial transactions and ensure that incomplete or invalid data is not permanently stored in the database.

Data consistency is another key aspect of reliability. PropSight is evaluated on whether financial records remain accurate and synchronized across all modules, even after multiple updates or user interactions. This ensures that income, expenses, and balances reflect the correct and most recent information at all times.

On top of this, the system is assessed in terms of recovery capability, particularly its ability to recover from minor failures or interruptions without requiring complete system restart or data loss. This contributes to the overall robustness of PropSight as a financial management tool for rental property owners.

3.7.4 Security

Security evaluates the robustness of the mechanisms within PropSight that protect sensitive financial data from unauthorized access, modification, or disclosure, including user authentication strength, role-based access control effectiveness, and database integrity.

Security also assesses the effectiveness of the system's authentication mechanisms in verifying the identity of users before granting access to the platform. This includes evaluating password protection strength, session management, and login validation processes to ensure that only authorized users can access PropSight.

Beyond that, the system's authorization controls are evaluated to confirm that rolebased access restrictions are properly enforced. Property owners and tenants are provided with different levels of access, ensuring that users can only view and interact with data relevant to their roles. This prevents unauthorized access to sensitive financial information and administrative functions. .

Data protection measures are also considered in the evaluation of security. This includes verifying that financial records, tenant information, and transaction data stored in the database are safeguarded against unauthorized modification or deletion. The system is assessed on its ability to maintain data confidentiality and integrity throughout all operations.

Along with this, the system is reviewed for protection against common web-based vulnerabilities, such as SQL injection and unauthorized data manipulation.

Secure coding practices implemented in the backend, including input validation and the use of prepared statements, are evaluated to ensure that PropSight maintains a secure and stable operational environment.

3.7.5 Efficiency

Efficiency measures whether PropSight allows property owners to complete their financial management tasks with less time and effort compared to their current methods, through automation of calculations and report compilation, and quick access to key information through the dashboard.

Efficiency also evaluates the system’s ability to optimize workflow processes by minimizing redundant steps in financial management tasks. PropSight is assessed on how effectively it streamlines operations such as data entry, payment recording, and report generation, ensuring that users can complete these processes in a logical and time-efficient sequence.

Adding to this, system responsiveness is considered as part of efficiency, particularly how quickly the system processes user inputs and retrieves or displays relevant financial information. This includes evaluating dashboard loading times, report generation speed, and the responsiveness of transaction recording functions under normal usage conditions.

Efficiency also measures the reduction of manual effort required from property owners in managing rental operations. By automating calculations such as income totals, expense summaries, and outstanding balances, the system reduces the need for repetitive manual computations, thereby decreasing workload and minimizing the risk of human error.

What is more, the system's ability to handle multiple records and transactions without significant performance degradation is evaluated. This ensures that PropSight remains efficient even as the number of properties, tenants, and financial entries increases over time.

SUPPORTING DIAGRAMS

Data Flow Diagram (DFD – Level 0)

The Level 0 DFD provides a high-level view of how PropSight interacts with its two main external entities: the Property Owner (Admin) and the Tenant (User). At this level, the entire system is represented as a single process, and the diagram shows the types of data flowing between the external actors and the system.

The Property Owner inputs property details, unit information, financial data such as income and expense entries, maintenance updates, and report requests into the system. In return, the system provides financial reports, dashboard analytics, tenant payment statuses, and maintenance request updates. The tenant submits booking requests, maintenance requests, and account registration information. The system responds with booking status confirmations, payment records, unit availability information, and maintenance acknowledgments. All data passes through the central PropSight system, which processes bookings, manages financial records, handles maintenance requests, and stores all data in the database for reporting and analytics.

The Level 0 DFD also establishes the foundational boundary of the PropSight system by clearly defining what is considered inside the system and what exists outside of it. This boundary specification is important in ensuring that all internal processes remain focused on financial monitoring and property management functions without ambiguity in system scope.

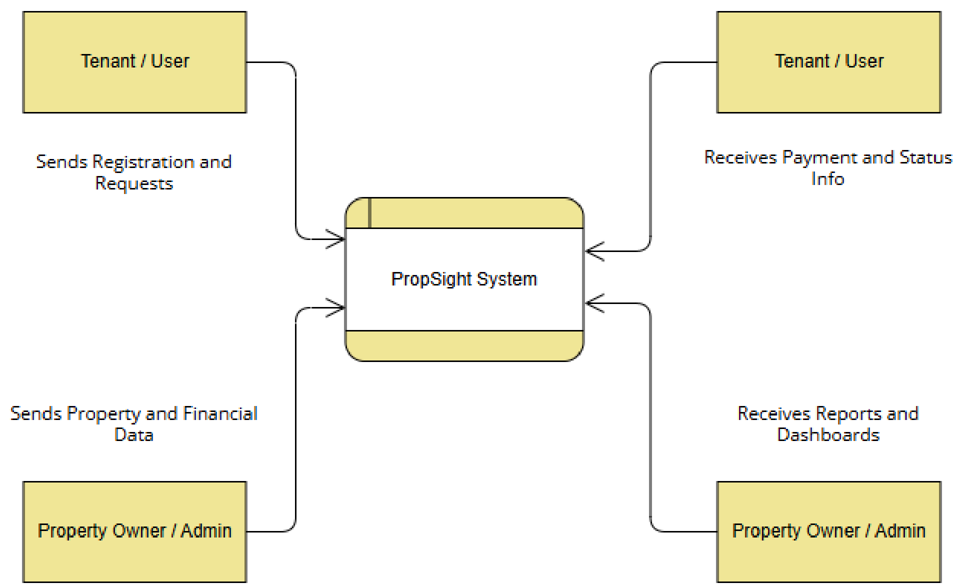
At the same time, this level emphasizes the central role of the PropSight system as an intermediary that processes all incoming data before generating meaningful outputs. All data inputs from both Property Owners and Tenants are validated, processed, and stored within the system before being transformed into useful information such as reports, dashboards, and status updates. This ensures consistency and accuracy in all system outputs.

The Level 0 DFD also highlights the bidirectional nature of communication between users and the system. Property Owners and Tenants are not only data providers but also recipients of processed information. This continuous exchange of input and output supports real-time interaction within the system and ensures that users remain updated with the latest financial and operational status.

Also, this level provides a conceptual foundation for the more detailed Level 1 and Level 2 DFDs, where each major process such as payment management, report generation, and maintenance handling is broken down into more specific subprocesses. This hierarchical decomposition ensures clarity in system design and supports systematic development and implementation of PropSight.

On the whole, the Level 0 DFD serves as a critical overview diagram that simplifies the system into its most essential interactions, allowing stakeholders to easily understand how PropSight functions as a centralized financial monitoring and property management system.

[DFD Level 0 Diagram - See Attached Figure]



Entity Relationship Diagram (ERD)

The ERD for PropSight defines the data entities managed by the system and the relationships between them, using standard ERD notation with primary keys (PK) and foreign keys (FK) identified for each entity.

In PropSight, the ERD typically includes key entities such as Users, Properties, Units, Tenants, Payments, Expenses, and Maintenance Requests. Each entity is designed to store specific sets of data relevant to rental property management, ensuring that all financial and operational information is properly structured within the database.

The relationships between entities are established to reflect real-world interactions within the rental property system. For example, a Property can contain multiple Units, a Tenant can be associated with a specific Unit, and each Tenant can generate multiple Payment records over time. Similarly, Properties are linked to multiple Expense records, and Maintenance Requests are associated with both Tenants and Units depending on the source of the request.

Primary keys (PK) are used to uniquely identify each record within an entity, ensuring that no duplicate entries exist in the database. Foreign keys (FK), on the other hand, are used to establish and enforce relationships between different entities, allowing data to be connected and retrieved accurately across multiple tables.

This structured relational design ensures data consistency, minimizes redundancy, and supports efficient querying of financial and property-related information. It also enables PropSight to generate accurate reports, track historical transactions, and maintain integrity across all system operations.

All in all, the ERD serves as the foundational blueprint of the database structure, ensuring that all components of PropSight are logically connected and aligned with the system’s financial monitoring and reporting objectives.

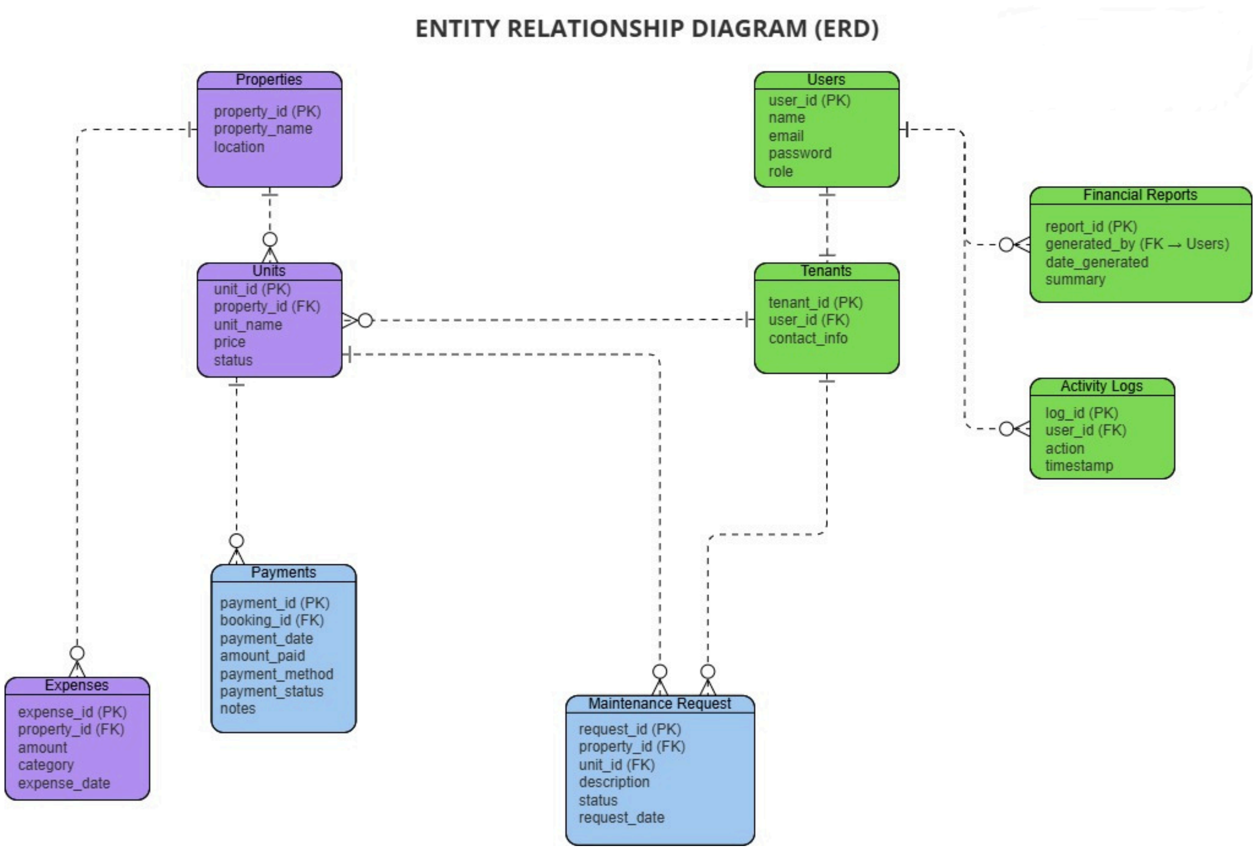
Entity	Attributes
Users	UserID (PK), Name, Email, Password, Role
Tenants	TenantID (PK), ContactInfo
Properties	PropertyID (PK), PropertyName, Location
Units	UnitID (PK), PropertyID (FK), TenantID (FK), UnitName, Price, Status
Payments	PaymentID (PK), BookingID (FK), Amount, PaymentDate, Status
Expenses	ExpenseID (PK), PropertyID (FK), Amount, Category, ExpenseDate
Maintenance Requests	RequestID (PK), TenantID (FK), UnitID (FK), Description, Status, RequestDate
Financial Reports	ReportID (PK), GeneratedBy (FK - Users), DateGenerated, Summary

Relationships

- A User can be linked to a Rental Record (1-to-Many between User and Rental Record).
- A Tenant can be linked to a Rental Record (1-to-Many between Tenant and Rental Record).
- A Unit can be linked to many Rental Records over time (1-to-Many between Unit and Rental Record).
- A Rental Record can have many Payments (1-to-Many between Rental Record and Payments).
- A User generates many Financial Reports (1-to-Many between User and FinancialReports).
- A Property contains many Units (1-to-Many between Property and Units).

- A Property has many Expenses (1-to-Many between Property and Expenses).
- A Unit has many Maintenance Requests (1-to-Many between Unit and MaintenanceRequests).
- A Tenant submits many Maintenance Requests (1-to-Many between Tenant and MaintenanceRequests).

[Entity Relationship Diagram - See Attached Figure]



Use Case Diagram

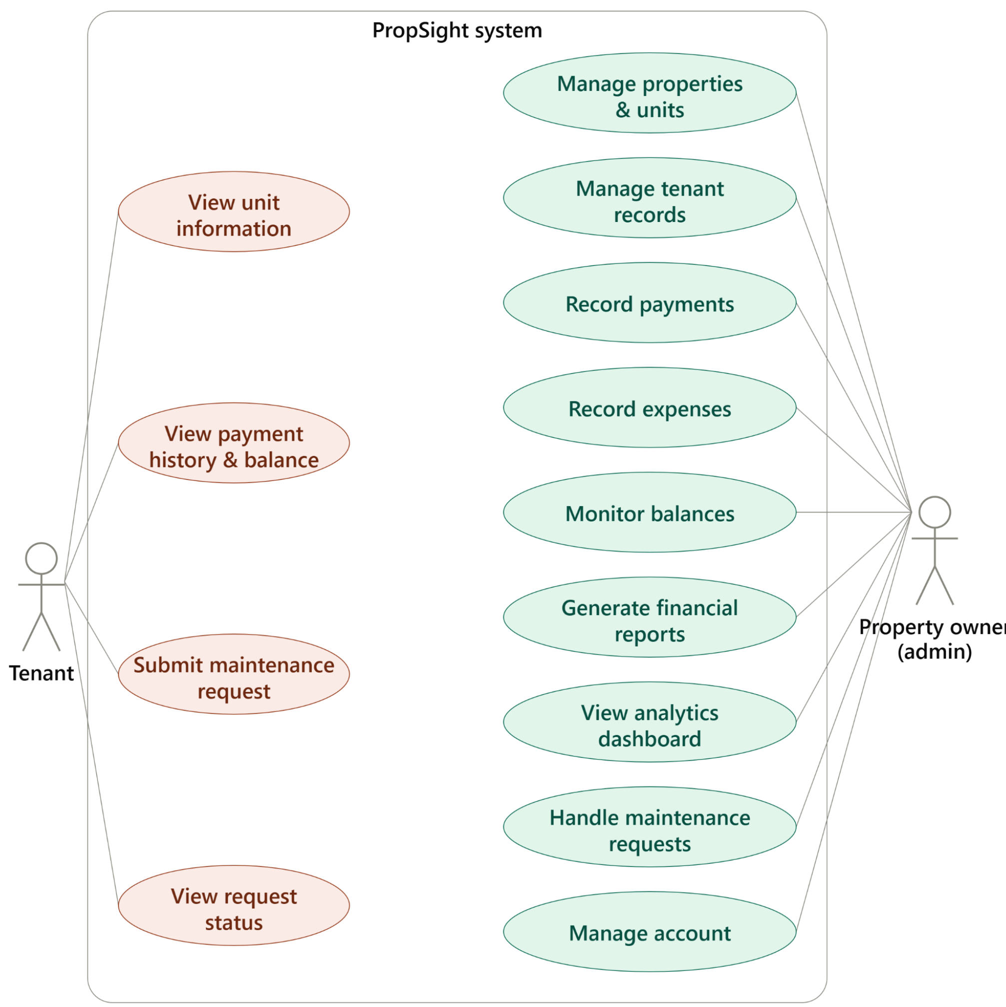
Use Cases for Property Owner (Admin)

- Manage Properties and Units: Add, edit, and monitor rental properties and the individual units within them, including unit status and rental price.
- Manage Tenant Records: Register tenants, update tenant profiles, and link tenants to the units they occupy.
- Record Payments: Enter rental payments received from tenants, with the system automatically updating outstanding balances.
- Record Expenses: Enter and categorize operational expenses associated with a property.
- Monitor Balances: Track which tenants have paid in full, paid partially, or have outstanding balances.
- Generate Financial Reports: Produce income summaries, expense breakdowns, and overall financial performance reports for a selected period.
- View Analytics Dashboard: Access charts and summary statistics on income, expenses, and account status.
- Handle Maintenance Requests: Log maintenance requests, link them to the relevant unit and tenant, and update their resolution status.
- Manage Account: Log in, log out, and manage basic account credentials.

Use Cases for Tenant (User)

- View Unit Information: View basic details of the unit they currently occupy.
- View Payment History and Balance: Check recorded past payments and the current outstanding balance.
- Submit Maintenance Request: Report a problem with the occupied unit for the property owner to review.
- View Maintenance Request Status: Check the status of a previously submitted maintenance request.

[User Case Diagram - See Attached Figure]



The supporting diagrams presented in this section serve as the primary visual communication tools for the architecture and behavior of PropSight. Each diagram type captures a different dimension of the system's design and provides a complementary perspective on how the system is structured and how it operates. Together, the DFD, ERD, and Use Case Diagram constitute a complete design specification that enables developers to understand the system's requirements, architects to understand its data structure, and users to understand the scope of its functionality.

The Data Flow Diagram at Level 0 provides the most abstract view of the system, treating PropSight as a single processing unit and focusing on the types of information that flow into and out of it. This high-level perspective is particularly useful for communicating the system's purpose and scope to stakeholders who are not involved in its technical development, such as the property owners who will use it, the academic faculty who will evaluate the capstone project, and future researchers who may wish to extend or adapt the system. The Level 0 DFD makes clear, at a glance, that PropSight serves two types of external actors (the Property Owner and the Tenant) and that the information flows between these actors and the system align with the system's stated objectives of financial monitoring, reporting, and operational management.

The Entity Relationship Diagram provides a more technically detailed view of the system, showing the specific data entities that PropSight manages and the relationships between them. The ERD is the primary reference document for the database design phase of PropSight's development, guiding the creation of the MySQL table schema and ensuring that all relationships between data entities are correctly represented and enforced through primary and foreign key constraints. The ERD also serves as a communication tool for the development team, providing a shared reference that ensures all team members have a consistent understanding of the system's data structure.

The normalization of the ERD entities deserves specific attention as a design decision. The PropSight database schema is designed to third normal form (3NF), which means that all non-key attributes in each table depend only on the primary key of that table and not on any other non-key attributes. This normalization eliminates redundancy in the data, ensures that updates to any piece of data need to be made only in one place, and reduces the risk of data inconsistencies arising from partial updates. While normalization can sometimes reduce query efficiency by requiring joins between tables, the scale of data expected in a small-scale rental property management system makes this trade-off entirely acceptable.

The Use Case Diagram provides a functional view of the system from the perspective of its users, identifying the specific operations that each type of user can perform. The Use Case Diagram is particularly valuable during the requirements analysis and planning phases of development, as it ensures that the development team has a complete and shared understanding of the system's intended functionality before development begins. It also serves as the basis for the test case development in the testing phase, with each use case generating one or more test cases that verify the system's ability to support the corresponding user operation correctly.

The Use Case Diagram for PropSight reveals an important asymmetry between the Admin and Tenant interfaces: the Admin has access to a much more comprehensive set of functions than the Tenant, reflecting the fundamentally asymmetric nature of the landlord-tenant relationship in the context of property management. The Admin is responsible for the full management of the property, including all financial recording, reporting, and strategic decision-making, while the Tenant's interactions with the system are limited to the specific self-service functions relevant to their role as an occupant of the property. This asymmetry is deliberately designed into the system's role-based access control architecture, ensuring that sensitive financial data and management functions remain accessible only to the authorized administrator.

CHAPTER 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.2 Overview of the Developed System

PropSight is a web-based financial monitoring and reporting system designed to help small- to medium-scale rental property owners in the Philippines record, monitor, and report the income and expenses associated with their rental operations. The system assists property owners in tracking tenant payments, categorizing operational expenses, and generating financial reports that summarize the overall performance of their properties.

Unlike general-purpose property management platforms that attempt to cover leasing, communication, and administrative tasks all at once, PropSight concentrates specifically on the financial side of rental property operations, consistent with the scope defined in Chapter 1. The application processes recorded transactions and generates a dashboard that presents financial summaries, charts, and account status indicators.

The system aims to provide small- to medium-scale property owners with an accessible, centralized alternative to the handwritten logbooks, carbon-copy receipts, and disconnected spreadsheets described in Chapter 1, thereby reducing manual errors and improving the transparency of rental financial records.

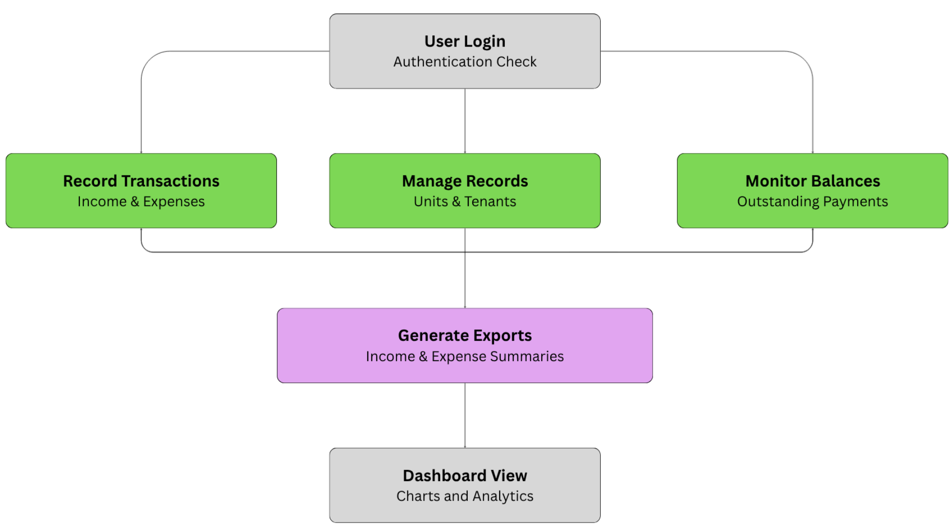
The major objectives of the developed system are as follows:

- Develop a centralized platform for recording and managing rental financial data.
- Generate accurate and timely financial reports for property owners.
- Provide real-time monitoring of rental payments and financial activities.
- Organize and present financial data in a structured, analytical format.
- Minimize errors, inconsistencies, and lack of transparency in manual financial management.

The development of PropSight follows modern software engineering principles by incorporating responsive interface design, secure database management, user authentication, and structured financial data processing.

The completed system consists of several interconnected modules that work together to capture, process, organize, and present a property owner's financial data.

Figure 4.1 illustrates the overall workflow of the developed system.



The workflow begins with user authentication. Once authenticated, property owners may record transactions, manage property and tenant information, monitor outstanding balances, and generate financial reports drawn from accumulated transaction data. The resulting information is presented through the dashboard using charts and structured summaries to facilitate interpretation by property owners.

4.3 System Development Methodology

The development of PropSight adopted the Agile Software Development Methodology, an iterative approach that emphasizes continuous collaboration, incremental development, and regular evaluation throughout the software development life cycle. Agile was selected because it accommodates evolving user requirements and promotes frequent feedback from stakeholders, helping ensure that the resulting application aligns with the operational needs of small-scale property owners.

The Agile methodology enabled the development team to divide the project into manageable iterations, commonly referred to as sprints. At the end of each sprint, a functional component of the system was produced, tested, and reviewed before proceeding to the next iteration. This process reduced development risks and supported early detection of software defects.

The development process consisted of the following phases:

Requirements Gathering

The first phase involved collecting information from prospective users through informal interviews and discussions with small-scale property owners. These consultations were used to determine the common challenges encountered in tracking rental income, monitoring tenant payments, and managing operational expenses. The gathered requirements served as the basis for defining the functional and non-functional requirements presented in Chapter 1.

System Planning

After identifying the project requirements, the researchers established the project scope, objectives, system specifications, development schedule, and resource allocation. The planning stage also considered the delimitations of the study, ensuring that the proposed features remained achievable within the timeframe and resources available for the project.

System Design

During this phase, the overall architecture of the application was designed. Database structures, process flows, user interface layouts, and system modules were prepared. The data flow diagrams, entity relationship diagram, and use case diagram presented in the Supporting Diagrams section of this paper were developed during this phase to provide a blueprint for software construction.

Software Development

The coding phase involved implementing the designed modules using the tools and technologies identified in Section 3.4. Individual system functionalities were developed incrementally, beginning with user authentication and the database schema, followed by property and unit management, tenant management, payment recording, expense tracking, and report generation.

Testing

Each completed module underwent unit testing before being integrated into the entire system. Integration testing ensured that the interaction between modules functioned correctly, while system testing verified the overall performance of the application. User Acceptance Testing (UAT) was later conducted with selected respondents to validate whether the application met user expectations

Deployment

As of the completion of this study, PropSight has been deployed to a local development and testing environment rather than a live production server. The system was hosted using XAMPP, which provided the Apache web server and MySQL database engine described in Section 4.5. The database schema was created and populated with sample property, tenant, and transaction records to support testing, with the application accessible only to the researchers and selected respondents during this phase.

Deployment to a live hosting environment is planned as an immediate next step following the completion of this study. This will involve migrating the application to a production web server, configuring a public domain, and applying the additional security measures identified in the recommendations in Chapter 5, such as stronger authentication, before the system is made available to actual property owners.

Maintenance

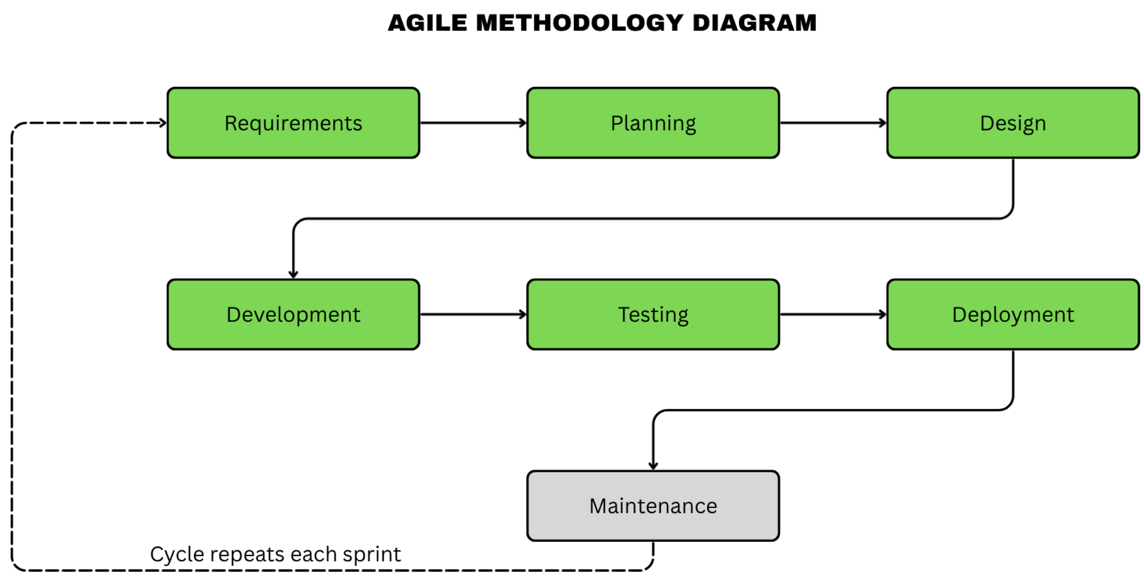
Maintenance represents the final phase of the Agile Software Development Methodology. After deployment, the researchers continuously monitored the performance of PropSight to identify software issues requiring correction or enhancement.

Maintenance activities include:

- Correcting software bugs
- Improving system performance
- Updating security features
- Enhancing user interface components
- Optimizing database performance
- Implementing feature improvements based on user feedback

Regular maintenance ensures that the developed application remains reliable, secure, and capable of adapting to future user requirements and technological advancements.

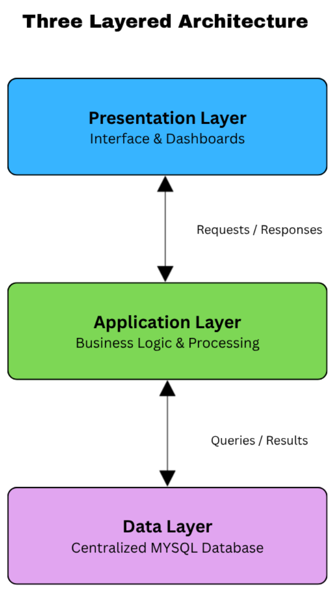
Figure 4.2 presents the Agile Software Development Life Cycle adopted in the study.



4.4 System Architecture

PropSight utilizes a Three-Layered Client-Server Architecture, which separates the presentation layer, application layer, and data layer. This architectural approach improves system maintainability, scalability, and security by organizing software components according to their respective responsibilities

Figure 4.3 illustrates the architecture of the developed system.



The architecture consists of the following components:

Presentation Layer

The presentation layer serves as the interface between property owners and the application. It is responsible for displaying information, receiving user input, and presenting financial reports through the dashboard and its accompanying visualizations.

Property owners may access the application using desktop computers, laptops, tablets, or smartphones through a modern web browser.

Major interface components include:

- Login Page
- Dashboard
- Property and Unit Management
- Tenant Management
- Payment Recording
- Expense Tracking
- Financial Reports

This layer emphasizes simplicity, responsiveness, and ease of navigation to improve user experience.

Application Layer

The application layer contains the business logic responsible for processing user requests, validating input data, calculating balances, and generating reports.

Its major responsibilities include:

- User Authentication
- Transaction Processing
- Tenant and Unit Management
- Balance Calculation
- Expense Categorization
- Report Generation

The application server acts as the intermediary between the presentation layer and the database server

Data Layer

The data layer stores all financial and property information in a centralized relational database. It is responsible for maintaining data integrity, enforcing relationships, and supporting efficient retrieval of information required by the reporting and dashboard modules.

Stored information includes:

- User Accounts
- Tenants
- Properties and Units
- Payments
- Expenses
- Maintenance Requests
- Financial Reports

The architecture ensures secure communication between the application server and the database while restricting unauthorized access to sensitive financial information.

Relationships among database tables are enforced through primary keys and foreign key constraints to ensure consistency and reliability. Proper indexing techniques were also implemented to improve query performance and support efficient generation of financial reports.

The Three-Layered Client–Server Architecture enables PropSight to securely process financial information while providing reliable access to authorized users. This architectural design also facilitates future expansion of the system, allowing additional modules and analytical features to be integrated without significantly modifying the existing framework.

4.5 Hardware and Software Requirements

The successful implementation of PropSight requires appropriate hardware and software resources. These specifications ensure that the application performs efficiently during data entry, balance calculation, and report generation.

4.5.1 Hardware Requirements

Table 4.1 presents the recommended hardware specifications

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3	Intel Core i5 or higher
Memory (RAM)	4 GB	8 GB or higher
Storage	256 GB SSD	512 GB SSD
Monitor	1366 x 768 Resolution	Full HD Display
Internet Connection	10 Mbps	50 Mbps
Mobile Device	Android 10	Android 13 or higher

The recommended hardware configuration provides sufficient processing capability for running the local development server and rendering the dashboard without significant delays.

4.5.2 Software Requirements

Table 4.2 presents the software requirements.

Software	Purpose
Windows 11 / Ubuntu Linux	Operating System
Visual Studio Code	Source Code Editor
MySQL	Database Management System
PHP	Backend Development
HTML / CSS / JavaScript	Frontend Development
Google Chrome	Web Browser
Git	Version Control
XAMPP	Local Development Server

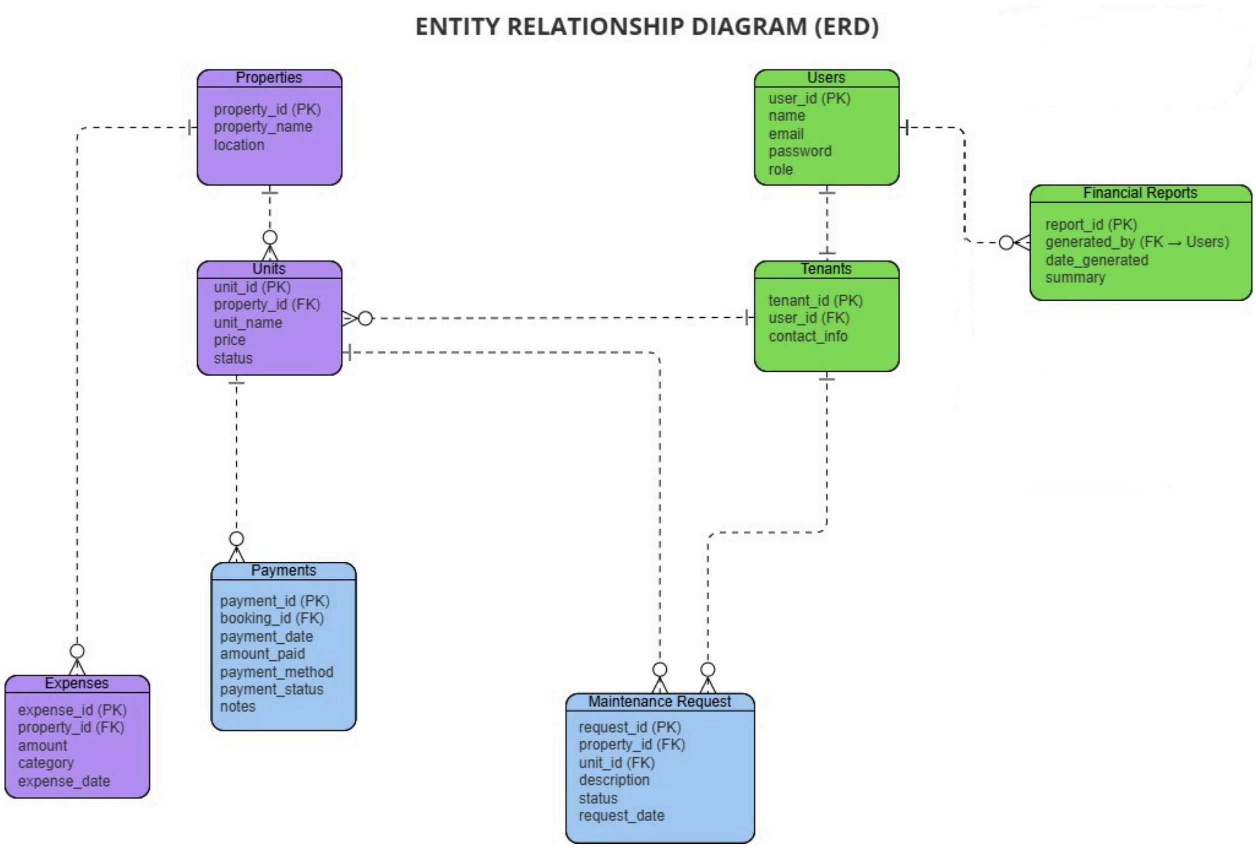
The selection of these software tools was based on their stability, community support, compatibility, and widespread adoption in modern web application development, consistent with the discussion in Section 3.4.

4.6 Database Design

The database serves as the foundation of PropSight by organizing and storing tenant records, property and unit information, payment and expense transactions, and generated reports. A well-designed database ensures efficient data retrieval, minimizes redundancy, and maintains data integrity throughout the operation of the system.

The researchers employed a relational database model using MySQL. Database normalization techniques were applied to reduce duplication and maintain consistency among related tables.

Figure 4.4 presents the Entity Relationship Diagram (ERD) of the system.



The primary entities of the database include:

- Users
- Properties
- Tenants
- Maintenance Request
- Payments

- Expenses
- Financial Reports
- Units

Users Table

This table stores account credentials and user information.

Field	Description
user_id	Primary Key
first_name	First Name
last_name	Last Name
email	User Email Address
password	Encrypted Password
role	User Access Level

Tenants Table

This table contains tenant information and links each tenant to the unit they occupy.

Field	Description
tenant_id	Primary Key
user_id	Foreign Key
contact_info	Tenant Contact Information

Properties Table

This table stores information about each registered property

Field	Description
property_id	Primary Key
property_name	Property Name
location	Property Location

Units Table

This table stores information about the individual units within a property.

Field	Description
unit_id	Primary Key
property_id	Foreign Key
tenant_id	Foreign Key
unit_name	Unit Name
price	Rental Price
status	Occupancy Status

Rental Payments Table

This table records rental payments made by tenants.

Field	Description
payment_id	Primary Key
tenant_id	Foreign Key
unit_id	Foreign Key
amount	Payment Amount
payment_date	Date Paid
status	Payment Status

Expenses Table

This table records operational expenses associated with a property.

Field	Description
expense_id	Primary Key
property_id	Foreign Key
amount	Expense Amount
category	Expense Category
expense_date	Date Incurred

Maintenance Requests Table

This table logs maintenance requests linked to a tenant and unit.

Field	Description
request_id	Primary Key
tenant_id	Foreign Key
unit_id	Foreign Key

description	Request Description
status	Request Status
request_date	Date Filed

Financial Reports Table

This table stores metadata for financial reports generated within the system.

Field	Description
report_id	Primary Key
generated_by	Foreign Key (Users)
date_generated	Date of Generation
summary	Report Summary

The relationships among these tables allow the application to trace tenant payments, monitor operational expenses, calculate outstanding balances, and generate comprehensive financial reports. Proper indexing and foreign key constraints help maintain referential integrity and optimize query performance, ensuring that the system can efficiently support both day-to-day recording and financial reporting.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter provides a comprehensive conclusion to the development of PropSight. It synthesizes the project's results, assesses whether the research objectives were achieved, and offers recommendations for future enhancements and implementation strategies to further maximize the system's utility for small to medium-scale rental property owners.

The discussion in this chapter draws directly on the results presented in Chapter 4, connecting the testing and evaluation outcomes back to the problem originally described in Chapter 1: the reliance of small-scale property owners on manual, error-prone methods for tracking rental income and expenses. Each section below revisits a specific part of that problem and considers how the completed system addressed it.

The chapter is organized to move from a summary of what was found, through the conclusions those findings support, to concrete recommendations for improving and extending the system. This structure mirrors the way the study itself proceeded: from an identified problem, to a designed and tested solution, to a set of next steps for continued development.

5.2 Summary of Findings

The development and evaluation of PropSight, described in detail in Chapter 4, drew on testing across four stages, unit, integration, system, and user acceptance testing, along with an evaluation against five criteria: functionality, usability, reliability, security, and efficiency. Taken together, these results yielded findings aligned with the research objectives:

- The system successfully centralizes the recording of rental income and operational expenses, removing the need for the handwritten logbooks and disconnected spreadsheets described in Chapter 1.
- The reporting module generates income summaries, expense breakdowns, and overall performance reports that are consistent regardless of when or how often they are accessed
- The payment monitoring module updates tenant balances as transactions are recorded, giving property owners a current rather than delayed view of financial activity.
- The dashboard presents financial data through charts and summaries that respondents were able to interpret without extensive guidance during user acceptance testing.
- Testing across unit, integration, system, and user acceptance levels did not surface any defects serious enough to prevent the system from performing its intended functions.

- Security and efficiency, while adequate for the system's intended small-scale use, were identified as areas that would need further work before the system could be deployed at a larger scale.
- Usability feedback was generally positive, though it also pointed to specific, addressable points of confusion, such as the visibility of the report generation feature.

Read together, these findings indicate that the system's core financial tracking functions, recording, monitoring, and reporting, are working as intended, while also identifying a smaller set of refinements that would strengthen the system further, discussed in Section 5.4.

5.3 Conclusion

PropSight has proven to be a viable and effective solution for small- to medium-scale property owners seeking to organize their rental financial records. By transforming scattered, manual records into a centralized, structured system, PropSight empowers owners to make informed decisions regarding income tracking, expense management, and overall financial planning. The project successfully addressed the need for an affordable, accessible, and reliable financial monitoring tool, demonstrating that structured financial systems can be made available to small-scale property operations. The objectives set out in Chapter 1 were substantially achieved, confirming that the system improves the accuracy and transparency of rental property financial management.

It is worth emphasizing that this conclusion rests on evidence gathered across all four stages of testing described in Chapter 4, not on a single measure of success. Unit and integration testing confirmed that the system's individual components work correctly, system testing confirmed that they work correctly together, and user acceptance testing confirmed that actual respondents were able to use the finished system to accomplish the tasks it was designed for.

At the same time, the study's delimitations remain relevant to how these findings should be read. PropSight was evaluated within a controlled academic environment and with a limited group of respondents, and its data accuracy still depends on correct data entry by its users. As such, the conclusions drawn here describe the system's performance under the conditions tested rather than under full-scale, long-term operational use.

Within those boundaries, the study demonstrates that a centralized, web-based system can meaningfully improve on the manual methods that small-scale property owners currently rely on. By reducing reliance on handwritten records and scattered spreadsheets, PropSight offers a practical, accessible alternative that does not require the cost or complexity of enterprise-level property management software.

The study's deliberate focus on financial monitoring and reporting, rather than the full range of features found in commercial property management platforms, also appears to have been a reasonable design choice. This narrower scope kept the system manageable to design, build, and test within the timeframe of a capstone project, while still addressing a problem that Chapter 1 identified as a genuine, recurring difficulty for small-scale property owners.

Table 5.1 maps each specific objective from Chapter 1 to the corresponding finding discussed in this chapter, providing a concise, at-a-glance view of how the study's original aims were addressed.

Specific Objective	Corresponding Finding
Centralized recording of financial data	Property and unit management, tenant management, and expense tracking modules consolidate records into one platform.
Accurate and timely financial reports	Financial report generation module produces consistent income, expense, and performance summaries.
Real-time monitoring of payments	Payment recording module updates tenant balances immediately upon entry.
Structured, analytical presentation of data	Dashboard module presents figures and trends through charts and summaries
Reduction of manual errors and inconsistency	Automated balance calculation, categorized expense entry, and activity logging reduce common manual error sources.

5.4 Recommendations

To ensure the continued relevance and efficacy of PropSight, the following recommendations are proposed, organized into system improvements, directions for future research, and strategies for property owners who plan to use the system as it currently stands.

These recommendations are grounded directly in the testing and evaluation results discussed in Chapter 4, rather than in speculative feature ideas, so that any future work builds on evidence gathered during this study rather than assumptions about what property owners might want.

System Improvements

The following improvements are recommended to address the gaps identified during testing and evaluation in Chapter 4, and to extend the system's usefulness beyond its current scope.

- Add automated notifications for upcoming or overdue tenant payments, so that property owners do not need to check the dashboard manually to catch late payments.
- Integrate an online payment feature to allow tenants to make rental payments securely through the system. This feature provides tenants with a more convenient and efficient way to settle their rental obligations without the need for manual payment transactions. It also helps property owners receive payments more quickly, reduces delays, minimises human errors in recording transactions, and automatically updates payment records within the system for improved accuracy and financial monitoring.
- Enhance security by implementing two-factor authentication and regular data backup to better protect users' financial information. Two-factor authentication provides an additional layer of protection by requiring users to verify their identity before accessing their accounts. Regular data backup ensures that important financial records and user information remain protected against accidental loss,

hardware failure, or cyber threats, allowing data to be restored when necessary and improving the overall reliability and security of the system.

- Develop additional reporting and export features that allow financial reports to be generated in multiple formats, such as PDF, Excel, and CSV, for easier record-keeping and sharing. These features provide property owners with greater flexibility in managing and analysing their financial information. Supporting multiple export formats makes it easier to store records, share reports with accountants or business partners, and maintain proper documentation for future reference. They also improve the accessibility and usability of financial data for decision-making and reporting purposes.

Future Research Directions

Beyond direct improvements to the system itself, the following research directions are recommended for those looking to build on the findings of this study

- Explore the integration of simple forecasting features to enable predictive analytics, such as projecting expected income based on historical payment patterns.
- Conduct a longitudinal study to analyze the long-term impact of the system on financial record accuracy and time savings for property owners.
- Investigate the feasibility of a mobile-native application to provide greater accessibility for property owners who primarily use mobile devices.

- Study the feasibility of integrating the system with common accounting or bookkeeping software, so that data recorded in PropSight can be reconciled with a property owner's broader financial records.
- Explore the integration of additional property management features, such as lease contract management, maintenance request tracking, tenant notification services, and document management, while preserving the system's primary focus on financial monitoring and reporting. Incorporating these features enables future versions of PropSight to provide a more comprehensive rental property management solution by supporting both financial and administrative tasks. These enhancements also improve operational efficiency, reduce manual processes, and provide property owners with a centralized platform for managing various aspects of their rental properties.
- Investigate the possibility of developing a mobile application that enables property owners to access the system through smartphones and tablets. A mobile version allows users to monitor rental income, record expenses, track tenant payments, generate financial reports, and review financial records anytime and anywhere. Improved accessibility increases user convenience, supports timely decision-making, and enables property owners to manage their rental properties even when they are away from their offices or business locations.

Implementation Strategies

For property owners who intend to begin using PropSight in its current form, the following practices are recommended to get the most reliable results from the system.

- Property owners should conduct a brief onboarding session for staff to ensure consistent data entry habits, which are critical for accurate financial reporting.
- Property owners are encouraged to perform regular data backups and system maintenance to ensure security and prevent data loss.
- Property owners managing multiple properties should register each property separately within the system from the outset, rather than consolidating unrelated properties under a single record, to keep reports accurate and easy to interpret.
- Conduct user orientation and training to ensure that property owners understand the features, functions, and proper use of PropSight. Training sessions should introduce users to the system's modules, including tenant management, financial recording, payment tracking, expense management, dashboard analytics, and report generation. Providing adequate training helps users transition from manual methods to a digital platform more effectively and reduces the likelihood of incorrect system usage. Well-informed users are more capable of maximizing the benefits of the system and maintaining accurate financial records.
- Implement PropSight through pilot testing before full deployment to validate system functionality and identify areas for improvement. Pilot implementation allows property owners to use the system in an actual rental management environment and provides an opportunity to evaluate its performance using real

operational data. Through this process, system developers can identify technical issues, assess usability, and gather feedback from users regarding system features and functionality. The results of pilot testing may serve as a basis for refining the system before wider implementation.

- Perform regular system maintenance and database backups to maintain system reliability, security, and data integrity. Routine maintenance activities, such as software updates, database optimization, and correction of technical issues, help ensure that the system continues to operate efficiently over time. Regular data backup procedures are also essential to protect financial information from accidental loss, hardware failure, or unexpected system interruptions. Maintaining updated and secure records supports the long-term reliability and sustainability of the system.

- Monitor system performance continuously to ensure efficient operation and promptly address technical issues. Continuous monitoring allows administrators and developers to evaluate system responsiveness, detect errors, and identify areas that require improvement. Monitoring also helps ensure that financial records, reports, and dashboard information remain accurate and current. Prompt resolution of technical concerns contributes to user satisfaction and prevents disruptions in financial monitoring and reporting activities.

- Collect user feedback regularly to support continuous system improvement and future feature enhancements. Feedback from property owners and users provides valuable insights regarding system usability, functionality, and overall effectiveness. Suggestions and comments from users may help identify additional requirements or features that could improve the system in future versions. Incorporating user feedback also supports the principles of continuous improvement and ensures that the system remains aligned with the needs of its intended users.

- Encourage accurate and consistent data entry to maintain reliable financial records and reports. Since the effectiveness of PropSight depends on the quality of information entered into the system, users should follow proper procedures in recording rental income, expenses, tenant payments, and other financial transactions. Consistent and accurate data entry reduces errors, improves report reliability, and supports effective financial monitoring. Maintaining complete and organized records also strengthens financial transparency and enables more informed decision-making.
- Evaluate the effectiveness of PropSight periodically to ensure that it continues to support efficient financial monitoring and reporting for rental property management. Periodic evaluation allows property owners and developers to assess whether the system continues to meet operational requirements and user expectations. Evaluations may include reviewing system performance, measuring user satisfaction, and identifying areas requiring enhancement. Continuous assessment ensures that the system remains relevant, effective, and adaptable to the changing needs of rental property management and technology.

5.5 Impact of the Study

PropSight contributes to the IT field by demonstrating the practical application of structured financial data management in a resource-constrained environment. By providing small- to medium-scale property owners with accessible financial monitoring tools, this study bridges the gap between complex financial management systems and everyday small-scale rental operations. While the project faced limitations, such as its dependence on accurate user-entered data and its evaluation within a controlled academic environment, the results underscore the significant potential for technology to level the playing field for small property owners in an increasingly data-driven market.

The study also offers a reference point for future capstone projects that aim to serve small business owners who are often underserved by enterprise-oriented software. By keeping the system's requirements grounded in the actual practices of small-scale property owners, as described in Chapter 1, PropSight illustrates how targeted digital tools can support the broader shift among small businesses in the Philippines toward more organized, data-informed financial management.

For the researchers, the study also served as a practical exercise in applying the Agile methodology, system design techniques, and evaluation practices covered throughout the program to a real, self-contained problem, from initial requirements gathering through to testing and evaluation of a completed system.

5.6 Summary

In summary, PropSight represents a successful integration of software engineering principles and structured financial data management. It serves as a demonstration of the value of tailored technology solutions in addressing real-world challenges faced by small-scale rental property owners. This research provides a solid foundation for future developers to build upon, contributing to the broader goal of accessible digital financial tools for small property owners in the Philippines.

The findings, conclusions, and recommendations presented in this chapter close the discussion of the completed system while pointing toward practical next steps for its continued development. It is hoped that this study provides a useful foundation for future work on accessible financial management tools for small property owners in the Philippines.

Ultimately, the value of PropSight lies not only in the specific features it implements, but in what it demonstrates about the feasibility of building focused, affordable financial tools for a segment of small business owners who are frequently overlooked by mainstream software vendors. That demonstration, more than any single feature, is the study's lasting contribution.

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